

Critical Care Medicine & Toxicology

Saturday, November 29, 2025

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BONUS

Read on own

Acronyms Used in this Talk

- ABG – arterial blood gas
- Abs - antibodies
- ACS-Acute coronary syndrome
- AE - adverse events
- AG – anion gap
- AGMP-Aerosol generating medical procedure
- AKI-Acute kidney injury
- ARDS-Acute Respiratory Distress Syndrome
- A&O-Alert and oriented
- AVM – arteriovenous malformation
- b/l-Baseline
- BIPAP-Bilevel positive airway pressure
- BG-Blood glucose
- BP-Blood pressure
- CAM – confusion assessment method
- CCCS – Canadian Critical Care Society
- CI - Contraindication
- CO-Cardiac output
- COT-Conventional oxygen therapy
- CPAP-Continuous positive airway pressure
- CVP-Central venous pressure
- Ecmo-Extracorporeal Membrane Oxygenation
- ED-Emergency department
- EPAP-Expiratory positive airway pressure
- ETOH - ethanol
- ETT-Endotracheal tube
- EVD – external ventricular drain
- FiO₂ – fraction of inspired oxygen
- FRC-Functional residual capacity
- GCS-Glasgow coma scale
- HFNC-High flow nasal cannula
- HR-Heart rate
- ICP – intracranial pressure
- ICU-Intensive care unit
- IFN - Interferon
- ILD – interstitial lung disease
- IPAP-Inspiratory positive airway pressure
- IMV-Invasive Mechanical Ventilation
- iNO-Inhaled nitric oxide
- IVC-Inferior vena cava
- I:E-Inspiratory Time to Expiratory Time Ratio
- JVP-Jugular venous pressure
- LOC-Level of consciousness
- LR-Likelihood ratio
- LV-Left ventricle
- MAP-Mean arterial pressure
- NAC - N-Acetylcysteine
- MEWS – Modified early warning score
- NEWS – National early warning score
- MG – myasthenia gravis
- NDD-Neurologic Determination of Death
- NIV-Non-invasive ventilation
- NMBA-Neuromuscular blocking agent
- NS-Normal Saline
- NYD-Not yet diagnosed
- OG – osmolar gap
- OSA – obstructive sleep apnea
- OHS – obesity hypoventilation syndrome
- Pplat-Plateau pressure
- PEEP-Positive end expiratory pressure
- PIP-Peak inspiratory pressure
- PPE-Personal protective equipment
- PSILI – patient self induced lung injury
- PSV-pressure support ventilation
- RASS – Richmond agitation-sedation scale
- RM – Rumack-Matthew
- RCT – randomized control trial
- RL-Ringer's lactate
- ROSC – return of spontaneous circulation
- RR-Respiratory Rate
- RSBI-Rapid shallow breathing index
- qSOFA-quick Sequential Organ Failure Assessment
- SBT-Spontaneous breathing trial
- SCC-Surviving Sepsis Campaign
- SIRS-Systemic Inflammatory Response Syndrome
- SUP-Stress Ulcer Prophylaxis
- SV-Stroke volume
- SVR-Systemic vascular resistance
- Sx-Surgery
- TFB – total fluid balance
- TCA-Tricyclic Antidepressant
- VBG – venous blood gas
- VF – ventricular fibrillation
- VT – ventricular tachycardia
- VTE-Venous thromboembolism
- Vt-Tidal volume
- VILI – ventilator induced lung injury
- V/Q-Ventilation/perfusion

OUTLINE

- Shock and Sepsis
- Respiratory failure
- Weaning from Mechanical Ventilation
- ICU Delirium, Sedation, Analgesia, Sleep
- Targeted temperature management, neuroprognostication, DNC, neuro ICU
- Toxicology
- Extra slides for your reference:
 - Acute hypoxia in the ICU
 - Gas trapping
 - Maternal cardiac arrest
 - Liver failure guidelines
 - Hyperthermia & Hypothermia
 - Toxicology Bonus Slides
 - Bonus MCQs

Useful Resources!

- <https://sccm.org/Home>
- <https://emcrit.org/category/emcrit/>
- <https://litfl.com/>
- <https://oncicu.com/Home>

MCQ 1 - 2026

A 57M with no past medical history is intubated in the emergency department for decreased level of consciousness. Initial CXR is unremarkable; urinalysis is positive with +3 leukocytes and +1 nitrites. He is found to be febrile at 39.1C and hypotensive at 87/34. After receiving 30cc/kg of crystalloid and IV piperacillin-tazobactam he is persistently hypotensive. Norepinephrine is initiated to maintain a MAP > 65mmHg. You are called to assess him 6 hours later for worsening hypotension. He is currently on 0.5mcg/kg/min of norepinephrine with MAP persistently 60-65mmHg. You perform a bedside POCUS demonstrating global hypokinesis of the left ventricle, a dilated IVC and B lines in the lungs.

What is your next best step?

- A. Increase upper limit of norepinephrine dosing to 1.0mcg/kg/min
- B. Initiate vasopressin at 2u/hr for MAP > 65mmHg
- C. Initiate dobutamine at 2.5mcg/kg/min
- D. Initiate milrinone at 0.5 mcg/kg/min
- E. Bolus 1L Ringer's Lactate and reassess for fluid responsiveness

Circulatory Shock Syndromes

Distributive	Hypovolemic	Cardiogenic	Obstructive
Loss of Vascular Tone	Low Pre-load	PUMP failure	Excessive After-load
- Sepsis - Anaphylaxis - SIRS (eg. pancreatitis) - Mitochondrial Dysf. (eg. cyanide) - Endocrine (thyroid, adrenal crisis) - Others*	- Hemorrhage - Trauma - Burns - Operative losses - GI losses - Renal losses - Third space losses (Pancreatitis)	- ACS - Arrhythmia - Acute Valvulopathy - Cardiomyopathy (restrictive or dilated) - Right or left sided heart failure	- Pulmonary Embolism - Tamponade - Tension pneumothorax
66%	16%	16%	2%

*Other Distributive: HLH, **medications**, liver failure, post cardiopulmonary bypass vasoplegia, neurogenic shock

Sepsis Defined (Surviving Sepsis Campaign, SCCM 2021)

“Life threatening organ dysfunction secondary to dysregulated host response to infection”

Use your clinical acumen, measure lactate, and a screening tool to diagnosis sepsis

Septic Shock = Sepsis +

- ✓ adequate volume resuscitation
- ✓ Persistent ↓BP requiring **vasopressors to keep MAP $\geq$ 65**
- ✓ **lactate > 2 mmol/L**

Screening Tools recommended by SCCM
SIRS, MEWS, NEWS (**not** qSOFA)

SIRS is easiest tool to memorise for oral exam :

HR >90

Temp >38 or <36

RR >20 or PaCO₂ <32

WBC >12 or <4 or 10% bands

2 or more = positive screen

LACTATE is helpful but cannot rely on exclusively

Sensitivity (83%)

Specificity (85%)

Positive LR (LR 4.75)

Negative LR (LR 0.29)



Sepsis: **IMMEDIATE** Initial Resuscitation

- If sepsis induced hypotension/shock, begin rapid administration of *at least* 30ml/kg (ideal body weight) crystalloid within the first 3h (2021 = Suggestion)
 - See next slide for guiding fluid resuscitation
- Use vasopressors if patient is hypotensive during or after fluid resuscitation to keep **MAP $\geq$ 65 ***
 - Norepinephrine (“levophed”) is first line vasopressor
 - * May target lower in some patient populations (elderly, etc).
- Give broad spectrum antibiotics **within 1 hour**
 - **Obtain blood cultures prior to antibiotics if possible**
- If ICU level care needed, admit to ICU within 6 hours

Diagnosis of Sepsis unclear?

DO : re-evaluate, perform diagnostic tests, can delay antibiotic start up to 3 h

DON'T : measure procalcitonin to help with decision to start antibiotics

1. Evans et al. Critical Care Medicine 2021; doi: 10.1097/CCM.0000000000005337

Sepsis: Antibiotic Choice

- Broad-spectrum antibiotics are recommended, and individual risk factors should be considered:
 - MRSA coverage:
 - High risk includes known MRSA colonisation, recurrent skin/soft tissue infections, PWID, central lines, dialysis.
 - Multi-drug-resistant (MDR) organisms – **double (2 antibiotic) Gram-negative coverage if high risk. **
 - High risk factors include recent antibiotics within 3 months, known MDR colonization, local prevalence, travel to endemic countries, or hospitalization abroad.
 - Fungal coverage should be considered if high risk:
 - High risk factors include neutropenia, immunocompromised status, TPN, dialysis, chronic lines, PWID, HIV, haematologic or solid organ transplants, emergency GI surgery, or anastomotic leak.
- Conduct daily assessments for de-escalation.
- Quickly determine if source control is needed (OR/IR/line removal).
- Prolonged beta-lactam infusion is associated with reduced 90-day mortality (high certainty).

1. SSC 2021. Intensive Care Med (2021) 47:1181–1247. doi.org/10.1007/s00134-021-06506-y
2. CCM (2021) 49:11. DOI: 10.1097/CCM.0000000000005357
3. Abdul-Aziz MH, Hammond NE, Brett SJ, et al. Prolonged vs Intermittent Infusions of β -Lactam Antibiotics in Adults With Sepsis or Septic Shock: A Systematic Review and Meta-Analysis. JAMA. 2024;332(8):638–648. doi:10.1001/jama.2024.9803 8

Sepsis: Guiding Resuscitation

We now recognize the harms of IVF/fluid intolerance. We titrate to **fluid responsiveness**.
Fluid responsiveness = $\uparrow$ stroke volume (SV) or cardiac output (CO) 10-15% after 500cc

- To guide resuscitation:
 - use dynamic variables > static parameters (weak, very low)
 - Trend lactate (weak, low)
 - Adjunct: capillary refill (<3 sec normal) (weak, low)
- **DYNAMIC Variables:**
 - Fluid bolus or Passive Leg Raise (45° raise x 30-90sec)
 - Pulse pressure variation (PPV) (>12%)
 - Echocardiography
 - Stroke volume or Stroke volume variation (SVV)
 - IVC
 - Intubated, fully ventilated- Distensibility Index (DI) >15-20%
→ likely to be fluid responsive
 - Intubated breathing spontaneously → cannot use IVC
 - Spontaneously breathing, not intubated, IVC <2cm and respiratory variation >50% → likely fluid responsive

PPV = equation tedious / not practical at bedside
DI = (Dmax - Dmin) / Dmean

$$PPV(\%) = \frac{PP_{\max} - PP_{\min}}{\frac{1}{2}(PP_{\max} + PP_{\min})} \times 100$$

WHICH FLUID?

✓ Recommend **Crystalloid** first line
✓ Suggest using balanced crystalloid over NS (SMART trial - $\downarrow$ major adverse kidney events at 30 days)

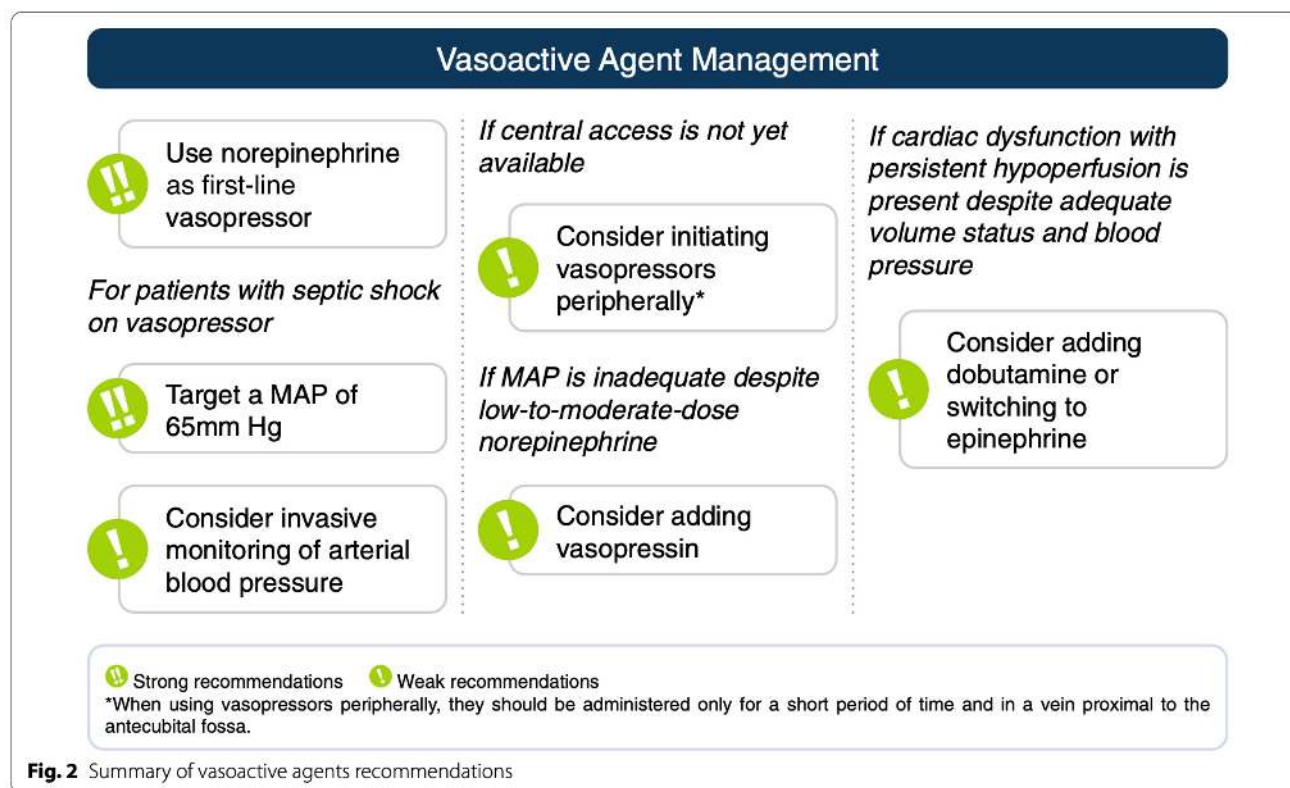
?Suggest albumin in patients who received large volumes of crystalloid over crystalloid alone

- Costly, no mortality benefit, did not define "large volume"; weak recommendation only

✗ Do not use starches (34 deaths per 1000)
✗ Do not use gelatin

1. Semler, M. et al. NEJM 2018; DOI: 10.1056/NEJMoa1711584
2. PETAL Network and NHLBI. NEJM 2023; 388:499-510: DOI: 10.1056/NEJMoa2212663

Hemodynamic Management and Vasopressors



Doses

Norepinephrine/Epinephrine
0.05-0.5mcg/kg/min
Vasopressin 2.4 units/hr
Dobutamine 2.5-10 mcg/kg/min

Add vasopressin when
Norepinephrine approx 0.25 –
0.5 mcg/kg/min

Consider MAP target 60-65 in elderly patients.

Trend towards possible mortality benefit with permissive hypotension in age > 65 and chronic hypertension.

Lamontagne F, et al. *JAMA* 2020; 323(10); 938-949

Fig. 2 Summary of vasoactive agents recommendations

BONUS

Read on own

BONUS: VASOPRESSOR PRIMER

Drug	Receptor stimulated	Best for...	HR	SVR	CO	PCWP
Phenylephrine	$\alpha 1$ +++	Septic Shock or during anesthesia	↓/-	↑	↓	↑
NorEpinephrine	$\alpha 1$ +++ $\beta 1$ ++	Septic shock FIRST LINE	↑	↑	↑/-	↑
Dopamine	Low dose: $\alpha 1$ 0 $\beta 1$ + $\beta 2$ 0 DA ++ High dose: $\alpha 1$ ++ $\beta 1$ ++ $\beta 2$ 0 DA ++	Septic shock (3 rd line) Caution re renal/mesenteric vasodilation at low dose, tachy arrhythmia at high dose	↑	↑	↑	↑
Epinephrine	$\alpha 1$ ++/+++ $\beta 1$ +++ $\beta 2$ ++	Anaphylactic shock Caution tachyarrhythmia at high dose	↑	↑	↑	↓
Dobutamine	$\alpha 1$ 0/+ $\beta 1$ +++ $\beta 2$ ++	Cardiogenic Shock *caution re risk of hypotension	↑	↓	↑	↓
Milrinone	PDE inhibitor	Cardiac surgery, Pulm HTN, Certain cardiogen shock	↑/-	↓	↑	↓
Vasopressin	V1 Rc on smooth muscle V2 Rc on Kidney	Septic Shock (2 nd Line)	↓/-	↑	↓/-	↑
Isoproterenol	$\alpha 1$ 0 $\beta 1$ +++ $\beta 2$ +++	Hypotension due to Bradyarrhythmia	↑↑	↓	↑	↓/-

Steroids and Adjuncts

Theory: Steroids may augment immune dysregulation, relative adrenal insufficiency

SCCM 2024 Suggest steroids for septic shock (conditional, low certainty)

- **Benefits:**

- ↓ Long-term mortality (*low certainty*)
- ↓ Short-term mortality (*moderate certainty*)
- ↓ ICU & hospital LOS (*low certainty*)

- **Risks:**

- ↑ Neuromuscular weakness (*low certainty*)
- ↑ Hypernatremia (*moderate certainty*)
- ↑ Neuropsychiatric complications (*low certainty*)

Effects on **GI bleed, MI, stroke, superinfection** → *uncertain*

 Practical Tips

- Typical dosing: **hydrocortisone 200 mg/day** ($\approx$ 50 mg IV q6h)
- **Avoid high dose** (400 mg/day) – *strong recommendation, moderate certainty*
- Consider when **norepinephrine ≥ 0.25 mcg/kg/min > 4 hrs**
- **Duration unclear** → caution if prolonged, may require taper

Do NOT give
(no proven benefit or may cause harm)

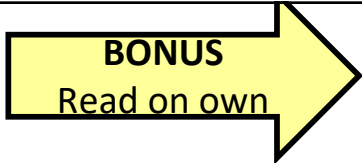
Immunoglobulins (IVIG)
Polymyxin
IV vitamin C
Angiotensin II infusion
Levosimendan
Activated Protein C
Liberal oxygen (target 94-96%, a sat >96% increases mortality in critically ill)



Best Practices in Sepsis: Summary of Guidelines

Screening	Hospitals to improve screening processes, SIRS/MEWS/NEWS > qSOFA, admit to ICU < 6hrs
Resuscitation	30ml/kg IBW balanced crystalloid, guide resuscitation with dynamic measures, CRT, lactate
Infection	Antibiotics in 1hr if probable or shock, cover resistant organisms if high risk, source control
Hemodynamics	MAP $\geq$ 65, Invasive BP monitoring, Norepi then Vaso, peripheral vasopressors for limited time
Steroids	Hydrocortisone 200mg/d when NE > 0.25mcg/kg/min for >4hrs
Ventilation	HFNC > NIV for Hypoxemic Resp Failure, LPV as per usual ARDS treatment (see slides on ARDS)
Best Practices	Restrictive Hb targets, VTE ppx, Stress ulcer prophylaxis, insulin for glucose > 10, Enteral feeding early
Long term outcomes	Address GOC early, integrate palliative care as needed, refer patient and family to support groups, involve social work, arrange follow up at discharge

CRT = Capillary Refill Time, Abx = antibiotics, NE = norepinephrine, Vaso = Vasopressin, HFNC = High flow nasal cannula, LPV = lung protective ventilation, ARDS = Acute Respiratory Distress Syndrome, NIV = Non-invasive ventilation, GOC = Goals of care, Hb = haemoglobin, VTE = Venous Thromboembolism



Parameter to ddx cardiogenic vs septic shock

(not common in clinical practice in 2025)

	Central Venous Gas (ScvO ₂)	Mixed Venous Gas (SvO ₂)
Draw from	IJ Central Line	Pulmonary Artery Catheter
Contains blood from	SVC	SVC and IVC
Normal O ₂ fraction	60-65%	65-70%

- ScvO₂ helps differentiate cardiogenic vs septic shock
- > 80% corresponds with high flow states (eg. sepsis). Mitochondrial dysfunction leads to reduced O₂ utilization
- < 65% indicates poor forward flow: cardiogenic shock
 - Here, the heart isn't strong enough to pump the oxygen out to the cells

Cardiogenic Shock

Low-cardiac-output state resulting in life-threatening end-organ hypoperfusion and hypoxia

- #1 cause is acute MI (STEMI >>NSTEMI)
- 1 year mortality 50% (largely unchanged last decade)

GIM Exam :

Make the diagnosis.
Initiate management,
transfer.

Best practices are for these
patients to be managed in
CVICU (outside GIM scope)

		Volume Status	
		Wet	Dry
Peripheral Circulation	Cold	Classic Cardiogenic Shock Cardiac index <2L/min/m ² . PCWP > 15 (↓CI; ↑SVRI; ↑PCWP)	Euvolemic Cardiogenic Shock (↓CI; ↑SVRI; ↔PCWP)
	Warm	Vasodilatory Cardiogenic Shock or Mixed Shock (↓CI; ↓/↔SVRI; ↑PCWP)	Vasodilatory Shock (Not Cardiogenic Shock) (↑CI; ↓SVRI; ↓PCWP)

Simplified Management:

- Volume Status
- Consider Effects of Positive Pressure Ventilation
- Vasopressor Support: Norepi generally used first
- Inotropes:
 - No strong evidence for preference of milrinone vs dobutamine. Considerations:
 - Dobutamine: quick onset (1-10 min), short half-life (2 min), chronic beta-blocker use can down regulate beta

receptors and make less effective

- Milrinone: onset (5-15 min), half-life (2.4 hours), Renally Cleared.

– Mechanical Circulatory Support (MCS)

- AHA recommends pts in persistent shock be evaluated for MCS candidacy by expert team
 - transfer to centre with this capability
- AHA recommends VA-ECMO if not expected to quickly improve with other devices

MCQ 1 - 2026

A 57M with no past medical history is intubated in the emergency department for decreased level of consciousness. Initial CXR is unremarkable; urinalysis is positive with +3 leukocytes and +1 nitrites. He is found to be febrile at 39.1C and hypotensive at 87/34. After receiving 30cc/kg of crystalloid and IV piperacillin-tazobactam he is persistently hypotensive. Norepinephrine is initiated to maintain a MAP > 65mmHg. You are called to assess him 6 hours later for worsening hypotension. He is currently on 0.5mcg/kg/min of norepinephrine with MAP persistently 60-65mmHg. You perform a bedside echocardiogram and find hypokineses of the left ventricle, a dilated IVC

What is your next best step?

- A. Increase upper limit of norepinephrine dose
- B. Initiate vasopressin at 2u/hr for MAP > 65
- C. **Initiate dobutamine at 2.5mcg/kg/min**
- D. Initiate milrinone at 0.5 mcg/kg/min
- E. Bolus 1L Ringer's Lactate and reassess for

C is correct. This is a scenario of likely sepsis-induced cardiomyopathy in the context of urosepsis. As the patient is progressively going up on his vasopressor requirements, he would benefit from an increase inotropy before adding another vasopressive agent (like vasopressin). Despite decreasing SVR, dobutamine will often result in an increase in MAP in these scenarios, given improvement in CO. Given his plethoric IVC and decreased LVEF, a fluid bolus is likely not going to benefit and may harm. Milrinone is not ideal in sepsis as it more significantly reduces SVR and may worsen hypotension in this patient.

OUTLINE

- Shock and Sepsis
- Respiratory failure
 - HFNC, NIV, Invasive MV
 - ARDS
- Weaning from Mechanical Ventilation
- ICU Delirium, Sedation, Analgesia, Sleep
- Targeted temperature management, neuroprognostication, DNC, neuro ICU
- Toxicology
- Extra slides for your reference:
 - Acute hypoxia in the ICU
 - Gas trapping
 - Maternal cardiac arrest
 - Liver failure guidelines
 - Hyperthermia & Hypothermia
 - Toxicology Bonus Slides
 - Bonus MCQs

MCQ 2 - 2026

A 55F is intubated in the ED for hypoxemic respiratory failure and brought to the ICU. Initial CXR demonstrates bilateral, diffuse patchy opacities. Antibiotics are initiated. Initial ventilator settings are ACVC Vt 5mL/kg, PEEP 5, FiO₂ 0.4, with Pplat 25. Initial ABG shows PaO₂ of 100 and SpO₂ 92%. She stabilizes initially. You are called back to assess the patient the next morning for worsening. Her PaO₂ is now 50 with SpO₂ of 85%. Ventilator settings are now ACVC Vt 5mL/kg, PEEP 8, FiO₂ 0.90, with Pplat 28.

What is your next step?

- A. Assess patient's candidacy for V-V ECMO.
- B. Initiate hydrocortisone 50mg IV q6h.
- C. Paralyze and prone patient.
- D. Gradually increase PEEP to 20 and assess response.

BONUS

Read on own

High Flow Nasal Cannula (HFNC)

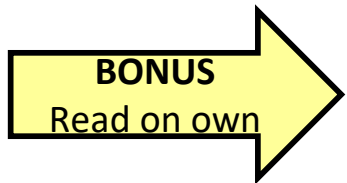
- Humidified O₂
- Flow up to 60 L/min
- FiO₂ up to 100%
- AGMP



Proposed Benefits:

Heated / Humidified Gas	Increased secretion clearance Decreased bronchoconstriction
Washout CO ₂ Upper Airways	Decreased dead space
High Nasal Insp. Flow	Decreased upper airway resistance
Positive Airway Pressure	Recruitment of atelectasis (gives minimal PEEP ~ 5 cmH ₂ O)
Decreased entrainment ambient air	Increased FiO ₂
Decreased work of breathing, transpulmonary pressures	Decreased patient self-induced lung injury.

Ozkowski et al. ERJ 2022;59:2101574.
Goligher et al. AJRCCM 2017;195(9).



HFNC in Acute Respiratory Failure

HFNC Should Probably be Used For:	HFNC Should Probably <u>Not</u> be Used For:
• Hypoxemic resp failure (adults): HFNC over Conventional O₂ Therapy (COT) or NIV • Patients taking Non-Invasive Ventilation (NIV) breaks: HFNC over COT • Post extubation (non-surgical patients) that are at a <u>low/mod</u> risk of extubation failure: HFNC over COT • Post-operative patient at <u>low risk</u> of respiratory complications: HFNC or COT • Post-operative patient at <u>high risk</u> of respiratory complications: HFNC or NIV	• Post extubation for patients at a <u>high risk</u> of extubation failure: NIV over HFNC (unless relative or absolute CI) • Acute hypercapnic resp failure secondary to COPD (pH < 7.35): trial NIV before HFNC

ERS Guidelines 2022

BONUS

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Non-Invasive Ventilation

Mode	CPAP – Continuous Positive Airway Pressure	BiPAP – Bi-Level Positive Airway Pressure
What does the machine provide?	Constant pressure throughout expiration and inspiration (i.e. PEEP or EPAP)	- Expiratory Positive Airway Pressure (EPAP = PEEP) AND - Inspiratory Positive Airway Pressure (IPAP)
Physiological Benefits	- Reduced respiratory muscle oxygen consumption - Recruitment of alveoli – Improved V/Q matching - Reduced LV afterload - Reduced Preload	- All the same as CPAP AND - Increased alveolar ventilation

NIV in Acute Respiratory Failure

NIV Should **Definitely** be Used For:

- BiPAP for mild-severe acidotic COPD patients (RR >20-24, pH ≤ 7.35, and PaCO₂ > 45)
 - BiPAP/CPAP for Cardiogenic pulmonary edema* (not cardiogenic shock and acute MI)
- Also in Pre-hospital setting (EMS)*

NIV Should Probably **Not** be Used For:

- Treatment of post-extubation resp failure
- Prevention of post-extubation resp failure if not high risk
- Hypercapnic COPD patients who are NOT acidotic

NIV Should **Probably** be Used For:

- Acute Respiratory Failure (ARF) if:
 - Post-operative patients (supra-diaphragmatic i.e. lung, abdominal, pelvic)
 - Chest trauma
 - Immunocompromised
 - Palliative patients if dyspneic from terminal cancers
- Prevention of post-extubation respiratory failure in high-risk patients (High risk: >65yo + underlying cardiac disease or respiratory disease)
- Weaning from invasive mechanical ventilation if hypercapnic resp failure

No Recommendation for:

- Asthma exacerbation
- De novo respiratory failure
- i.e., hypoxemia NYD, ARDS)
- Acute respiratory failure due to viral illness during pandemic

Contraindications to NIV

- Facial surgery, facial trauma, airway obstruction
- Decreased LOC (*relative)
- Inability to clear secretions
- Respiratory arrest
- Hemodynamic instability (reduces preload)
- **Other Indication for intubation**
(e.g. airway protection)

Postop Patients:

OK to proceed with NIV if indicated

(even if upper GI or thoracic surgery including esophageal - get blessing of surgeon if concern for anastomotic leak)

↓ reintubation ↓ invasive ventilation days

Intubating Evidence:

Setup for Success on 1st pass:

- Preoxygenate before intubation with NIV (not face mask) [PREOXI Trial, NEJM 2024]
- Video laryngoscope increases 1st pass rate [DEVICE trial, NEJM 2023]

Basic Modes of Ventilation

Mode	Type of Breath	What is Set (Independent Variable)	What Varies (Dependent Variable)	Best For	Risks / Limitations
Volume Assist/Control (A/C-VC)	In A/C the ventilator can deliver breaths either: Patient-triggered (assist) or Ventilator-triggered (control – i.e. apneic pt)	Tidal Volume (Vt), Rate, FiO ₂ , PEEP	Airway Pressures (Peak, Plateau)	Lung protection strategies (e.g., ARDS); Post-intubation (low GCS, apnea)	Volutrauma, breath stacking, auto-PEEP, patient-ventilator dyssynchrony
Pressure Assist/Control (A/C-PC)		Inspiratory Pressure, Rate, FiO ₂ , PEEP, I:E ratio	Tidal Volume (varies with compliance & resistance)	Poor lung compliance (ARDS), patient comfort	Hypoventilation if compliance worsens, variable minute ventilation, risk of hypoxemia
SIMV (Volume or Pressure)	Mandatory breaths + Spontaneous breaths allowed	Set tidal volume or inspiratory pressure (mandatory breaths), Rate, FiO ₂ , PEEP	Tidal volume of spontaneous breaths (patient effort ± PS)	Weaning (with PS), mixed control + spontaneous breathing	Can ↑ work of breathing if PS not added; less synchrony than A/C
Pressure Support Ventilation (PSV)	Supported (all patient-initiated)	Inspiratory Pressure, FiO ₂ , PEEP	Tidal Volume, Respiratory Rate (patient-dependent)	Patient comfort, weaning trials, spontaneous breathing support	Apnea if patient fatigues or becomes obtunded; variable tidal volumes

Ventilator Hacks

How to ↓ PaCO₂

↑ RR
↑ tidal volume
(minute ventilation = RR * Vt)

How to ↑ SpO₂
(PaO₂)

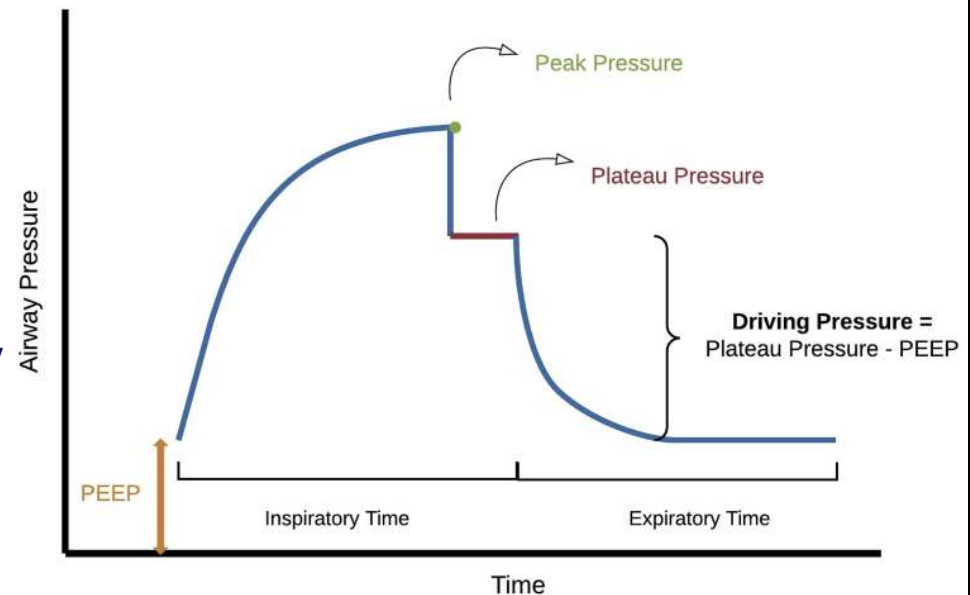
↑ FiO₂
↑ PEEP
↑ Inspiratory time
Affect O₂ delivery: ↑ cardiac output, ↑ Hb
↓ O₂ consumption: treat fever, agitation
Stop pulm vasodilators (eg nitroprusside)

Example of “Vent Orders”

RR 10-12, VT 6-8ml/kg (6 if ARDS)
PEEP 5-20, PC 5-25

Lung and Airway Pressures

- **Peak inspiratory pressures (PIP)**
 - Reflects airway resistance + lung compliance
 - Target <35cm H₂O
- **Plateau pressure (Pplat)**
 - The pressure in the lungs when no air is moving, reflects lung compliance
 - This is the pressure that the alveoli are seeing
 - Target <30 cm H₂O in any vented patient
 - >30 = risk of barotrauma causing pneumothorax/lung injury
 - May be confounded by chest wall restriction (ie. Obesity)
- **Positive End Expiratory Pressure (PEEP)**
 - Pressure at end of respiratory cycle
 - Increases solubility of gas (oxygen)
 - Splints airways, decreases work of breathing
- **Driving pressure:** Pplat – PEEP (target < 15)
OR tidal volume/static compliance



$$\text{Respiratory System Compliance} = \text{Tidal Volume} / (\text{Plateau Pressure} - \text{PEEP})$$

ARDS Diagnostic Criteria

	MILD	MODERATE	SEVERE
Timing	Within 1 week of known clinical insult or worsening respiratory symptoms		
Pulmonary edema	Not cardiogenic pulmonary edema or due to intravascular volume overload		
Chest Imaging	Bilateral infiltrates on CXR or CT Or on LUNG ULTRASOUND by trained professional *not explained by nodules, pleural effusions or atelectasis		
Hypoxemia PaO ₂ /FiO ₂ ** (**no requirement for PEEP in resource limited countries)	201-300 with NIV/CPAP PEEP ≥ 5 or HFNC >30l/min	101-200 PEEP ≥ 5	≤ 100 PEEP ≥ 5
Hypoxemia SpO ₂ /FiO ₂	≤ 315 with SpO ₂ $\leq 97\%$		

1. Riviello, Am J Respir Crit Care med 2016;193:52-9. 10.1164/rccm.201503-0584OC
2. Mathay et al., Am J Respir Crit Care Med, 2023, 10.1164/rccm.202303-0558WS.

ARDS Causes and Pathophysiology

DIRECT LUNG INJURY	SYSTEMIC INFLAMMATION
• Pneumonia • Aspiration pneumonitis • Drowning • Thoracic trauma/pulmonary contusion • Smoke or toxic inhalation • Fat emboli • Reperfusion injury (post lung transplant)	• Severe sepsis • Transfusion reaction (TRALI) • Shock • Pancreatitis

Pathophysiology:

- Proteinaceous fluid fills alveoli
- Neutrophils flood alveolar space
- Hyaline membranes form on epithelial basement membrane
- Microthrombi form
- Fibrosis develops (late stage)

Treatment:

1. Ventilation Strategies
2. Prone positioning
3. Neuromuscular Blockade
4. ECLS/ECMO
5. Corticosteroids
6. Inhaled Pulmonary Vasodilators

ARDS: Ventilation

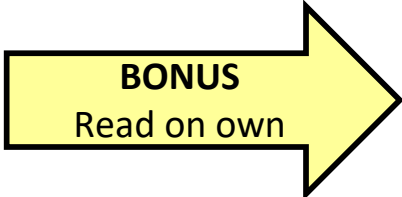
Patient Body weight =
Ideal Body weight
(your RT will adjust)

- **Mode:** Volume Control
- **Tidal Volume (Vt):** Start 6 mL/kg PBW → adjust 4–8 mL/kg PBW
- **Plateau Pressure:** ≤30 cm H₂O
- **Driving Pressure (Pplat – PEEP):** <15 cm H₂O
- **PEEP:** Use higher PEEP in moderate–severe ARDS (FiO₂–PEEP tables)
- **Oxygenation Target:** SpO₂ 88–93% *or* PaO₂ 55–80 mmHg (avoid hyperoxia)
- **CO₂ / pH:** Permissive hypercapnia acceptable; maintain pH >7.25
- **Adjuncts:**
 - Deep sedation may be required
 - **Lung recruitment maneuvers:** not routine (↑ mortality); selective use only
 - **High-frequency oscillation:** not recommended

FiO ₂	0.3-0.4	0.4	0.5	0.6	0.7	0.8	0.9
PEEP	5-8	8-14	8-16	10-20	10-20	14-22	16-22

ARDS Treatment Summary (ATS 2024, SCCM 2024)

	Severity of ARDS (P _a O ₂ /FiO ₂)		
	Mild (200-300)	Moderate (100-200)	Severe (<100)
Strong recommendation in favour	Lung protective ventilation (V _t 6 ml/kg PBW (range 4-8) + P _{plat} ≤30 cm H ₂ O		
		Prone Positioning >12h per day	
Conditional recommendation in favour	Systemic Corticosteroids Optimal regimen unknown. In DEXA-ARDS DEX 20mg IV x5d then 10mg IV x 5 d, started within 24-48h. Monitor for metabolic and immunosuppression risk. Starting after 14d of mech ventilation may be associated with harm.		
		High PEEP Caution if hemodynamically unstable or barotrauma risk	
			VV-ECMO (PaO ₂ /FiO ₂ <80)
			Neuromuscular blockade (use early, infusions <48h duration)
Strong recommendation against	High Frequency Oscillation Ventilation Prolonged recruitment maneuvers		



ECMO Respiratory Considerations beyond GIM scope

Consider ECMO	Do NOT Consider ECMO
- Severe ARDS - Hypercapneic respiratory failure - Bridge to lung transplantation - Primary graft dysfunction after lung transplantation - Status asthmaticus	<u>Absolute</u> - Disseminated malignancy - Known severe brain injury - Prolonged CPR without adequate tissue perfusion - Severe chronic organ dysfunction - Severe chronic pulmonary hypertension - Non-recoverable advanced comorbidity (ie. CNS damage or terminal malignancy)

Call for ECMO referral if:

- P/F < 80 mmHg for > 6 hours OR P/F < 50 mmHg for > 3 hours
- PaCO2 > 60 mmHg for > 6 hours (despite optimization of vent)
- Mechanically ventilated < 7 days
- BMI < 40 or weight < 125 kg
- Age 18 - 65

MCQ 2 - 2026

A 55F is intubated in the ED for hypoxemic respiratory failure and brought to the ICU. Initial CXR demonstrates bilateral, diffuse patchy opacities. Antibiotics are initiated. Initial ventilator settings are ACVC Vt 5mL/kg, PEEP 5, FiO₂ 0.4, with Pplat 25. Initial ABG shows PaO₂ of 100 and SpO₂ 92%. She stabilizes initially. You are called back to assess the patient the next morning for worsening. Her PaO₂ is now 50 with SpO₂ of 85%. Ventilator settings are now ACVC Vt 5mL/kg, PEEP 8, FiO₂ 0.90, with Pplat 28.

What is your next step?

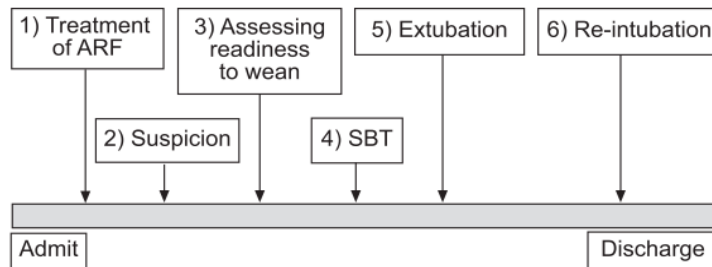
- A. Assess patient's candidacy for V-V ECMO.
- B. Initiate hydrocortisone 50mg IV q6h.
- C. Paralyze and prone patient.
- D. **Gradually increase PEEP to 20 and assess response.**

D is correct. While the other options may eventually be required, her PEEP is far too low for the severity of her ARDS (see PEEP table in ARDS slides). Before proning, we should optimize her ventilatory parameters to see if more aggressive intervention can be avoided.

OUTLINE

- Shock and Sepsis
- Respiratory failure
- **Weaning from Mechanical Ventilation**
- ICU Delirium, Sedation, Analgesia, Sleep
- Targeted temperature management, neuroprognostication, DNC, neuro ICU
- Toxicology
- Extra slides for your reference:
 - Acute hypoxia in the ICU
 - Gas trapping
 - Maternal cardiac arrest
 - Liver failure guidelines
 - Hyperthermia & Hypothermia
 - Toxicology Bonus Slides
 - Bonus MCQs

Weaning from Mechanical Ventilation



1. Boles et al., Eur Respir J 2007; 29: 1033–1056
DOI:10.1183/09031936.00010206
2. Fan et al., Ann Am Thorac Soc Vol 14, No 3, pp 441–443, Mar 2017. DOI: 10.1513/AnnalsATS.201612-993CME

Assess readiness for weaning from ventilator

1. Reversal of underlying reason for intubation and ventilation
2. Improvement of oxygenation ($\text{PaO}_2 > 60 \text{ mmHg}$, $\text{FiO}_2 < 40\%$, $\text{PEEP} < 8$)
3. Ability to perform work of breathing
 - Normal/compensated CO_2 , pH, adequate cardiac function, adequate diaphragm function)

Assess readiness for extubation

1. Adequate cough
2. Minimal secretions, ability to manage secretions
3. Awake/Alert, following commands, no sedation
4. No increased risk of airway obstruction – post-op swelling resolved, ETT cuff leak present

Spontaneous Breathing Trial (SBT)

- Mimics patient's own ability to breathe independently
- Perform on PSV for 30 minutes
 - PSV 5/5 vs. 0/0 vs. 5/0 → variable practices
 - Toronto standardized practice is 0/0
- Failure of SBT → tachypnea, increased work of breathing, tachycardia, hypotension, desaturation
 - Always ask 'why', and try to correct underlying cause
- Rapid shallow breathing index (RSBI) → $\frac{\text{Respiratory Rate (breaths/min)}}{V_t \text{ tidal volume (L/min)}}$
 - RSBI > 105 is a predictor of extubation failure

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Sedation

Sedation in Mechanically Ventilated Patients

- **Target:** RASS –2 to +1 (light sedation)
- **How:** Use daily sedation breaks and nurse-driven titration
- **Preferred agents:** Propofol or dexmedetomidine
 - **Benefits vs benzodiazepine infusion:** Shorter ICU stay, less delirium, shorter time to extubation
 - **Caution with dexmedetomidine:** Watch for bradycardia & hypotension; avoid dexmedetomidine if deep sedation is needed

Richmond Agitation Sedation Scale

4+	Combative, Violent, Dangerous
3+	Pulls/removes tubes or catheter, aggressive
2+	Frequent non-purposeful mvmt, dysynchronous
1+	Anxious, apprehensive, but not aggressive
0	Alert and calm
-1	Awakens to voice, eyes open > 10 seconds
-2	Lid sedation, briefly awakens < 10 seconds
-3	Moderate sedation, moves/eyes open
-4	Deep sedation, no response to voice, moves with physical stimulus
-5	Unroutable, no response to voice or stimulus

BONUS

Read on own

A Note on Dexmedetomidine

- Alpha-2 agonist – acts on CNS receptors
- Recent meta-analysis compared to Propofol and Benzodiazepines:
 - ↓ risk of delirium (RR 0.67, 95% CI 0.55 to 0.81; moderate certainty)
 - ↓ duration of mechanical ventilation (MD - 1.8 h, 95% CI - 2.89 to - 0.71; low certainty)
 - ↓ ICU length of stay (MD - 0.32 days, 95% CI - 0.42 to - 0.22; low certainty)
 - **Increased** risk of bradycardia (RR 2.39, 95% CI 1.82 to 3.13; moderate certainty) and hypotension (RR 1.32, 95% CI 1.07 to 1.63; low certainty)
- European 2024 Post-Op Delirium Guidelines:
 - Suggest dexmedetomidine for treatment of post-op delirium in cardiac surgery (weak recommendation, very low quality)

Lewis et al. Intensive Care Med 2022; 48(7): 811-840

Delirium (2018 SCCM Guidelines, 2024 Focused Update)

ICU Delirium Management

- **Non-drug:** Early mobility, promote sleep, optimize hearing/vision, frequent reorientation, reduce deliriogenic meds/transfusions.
- **Drug:**
 - Do **not** use antipsychotics or statins for prevention.
 - Avoid routine Haldol/atypicals—reserve for severe agitation, hallucinations, or distress.
 - Consider **dexmedetomidine** in ventilated patients to aid extubation.
- **Risk factors:**
 - *Modifiable* → benzodiazepines, transfusions
 - *Non-modifiable* → older age, dementia, prior coma, emergency surgery/trauma, high APACHE/ASA scores
- **Screen:** CAM-ICU
- **Outcomes of delirium:** ↑ cognitive impairment (3–12 mo), ↑ ICU length of stay

AID-ICU NEJM 2023 Anderson-Randberg

- Haloperidol vs placebo for delirium in ICU
- Number of days alive/out of hospital no different at 90d
- Number of rescue medications for agitation the same in both groups

Pain in the ICU (2018 SCCM Guidelines, 2024 Focused Update)

Multimodal therapy:

- **Opioids:** mainstay (esp. post-op) but cause resp. depression, delirium, dependence
- **Adjuncts to spare opioids:**
 - Acetaminophen, NSAIDs (if safe)
 - Low-dose ketamine (0.5 mg/kg bolus → 1–2 mcg/kg/min) post-op
 - Gabapentin/pregabalin/carbamazepine for neuropathic pain
- **Non-pharm:** cold therapy, relaxation, music, massage
- **Not routine:** lidocaine, local anesthetics, inhaled agents (special cases only)

Sleep (2024 SCCM Guidelines)

- **Non-pharm:** Reduce noise (earplugs), light (eye shades), and nighttime disruptions
- **Drugs:** Avoid propofol; melatonin **conditionally recommended** for sleep & delirium prevention; insufficient evidence for dexmedetomidine
- **Ventilated patients:** Consider assist-control (vs. pressure support) overnight

OUTLINE

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BONUS
Read on own

Neurologic Outcomes

- **Brain Death:**
 - Irreversible cessation of cerebral and brainstem function
- **Persistent Vegetative State:**
 - Severe anoxic brain injury progressing to a state of wakefulness without awareness
 - No purposeful responses, sleep wake cycles intact
- **Minimal Conscious State:**
 - Limited interaction with environment with visually tracking +/- simple commands
 - Intelligible verbalization or sometimes yes/no but not always appropriate
- **Locked In:**
 - Retained alertness, cognitive abilities, can move eyes and blink voluntarily, paralysis of the limbs and oral structures

BONUS
Read on own

Neuroprognostication

CPC	Disability	Conscious	Independent	Features
1	No, minor	Yes	Yes	Alert, able to work and lead a normal life with no/mild deficits.
2	Moderate	Yes	Yes	Independent in activities of daily life. May have hemiplegia, seizures, ataxia, memory impairment.
3	Severe	Yes	No	Limited cognition, dementia, locked in, minimally conscious. Usually requires institution level care.
4	Unconscious	No	No	Persistent vegetative state.
5	Dead	--	--	Certified brain death, circulation preserved

Cerebral Performance Categories. 1, 2 = Favourable outcome. 3, 4, 5 = Poor outcome

Summary of Neuroprognostication Guidelines

CJC 2023

Recommendation	Guideline: CJC 2023
Patient	Adults (>18) with ROSC, GCS < 8, unable to follow commands
Timing	Wait ≥24h post-arrest before prognostication
Neurologic exam	- Pupillary light reflex absent at 72h → poor outcome - Corneal reflex absent at 72h → helpful but residual sedation can confound - Motor exam unreliable (high false +) - Ensure adequate time for sedatives, opioids, paralytics to be cleared
Seizures/Status myoclonus	Status myoclonus < 7d after CA = marker of poor outcome when combined with other findings Treat status epilepticus as usual, delay neuroprognostication if prognosis favourable
Neurophysiological Testing SSEP = somatosensory evoked potentials	Use EEG as 1. part of multimodal assessment 2. to distinguish between SE and status myoclonus Use SSEPs as part of assessment. Bilaterally absent N20s > 24hrs after CA highly predictive of poor outcome Don't use electrographic status epilepticus or absence of EEG activity as a marker of a poor outcome
Serum biomarkers	Don't use routinely
Neuroimaging	CT ≤72hrs after ROSC. Reduced grey white matter differentiation = poor outcome MRI ≤ 7d after ROSC if uncertainty exists. Large diffusion restriction (high DWI) = poor outcome
Decisions	Use results from ≥2 different modalities before the decision to withdraw life-sustaining therapies (modalities = Neurologic exam, EEG, SSEPs, Neuroimaging, Serum biomarkers)

NOT HELPFUL:
Arrest rhythm
Patient's age
Arrest duration

Criteria for Determination of Death

- Death = **Permanent cessation of brain function**
 - Absence of consciousness
 - Absence of brainstem reflexes
- Death caused by **cessation of blood flow to brain**, due to:
 1. Circulatory Arrest – Death by Circulatory Criteria (DCC)
 2. Devastating brain injury – Death by Neurological Criteria (DNC)
- Donation can occur after:
 1. DNC
 2. DCC
 - Controlled circumstances – WLSM, MAiD
 - Uncontrolled circumstances – i.e. unanticipated arrest. NPOD

WLSM = withdrawal of life-sustaining measures, NPOD = non-perfused organ donation

Donation after Death by Circulatory Criteria (DCC)

Determining Death by Circulatory Criteria (DCC)

- **Principle:** No extracranial circulation → permanent absence of intracranial circulation
- **Monitoring:** Invasive arterial BP + continuous ECG only; **do not** use echo, palpation, or US
- **Criteria:** Pulse pressure <5 mmHg indicates absent circulation
- **Observation time:**
 - Controlled situation (WLSM/MAiD): 5 min
 - Uncontrolled situation: 10 min

Death by Neurologic Criteria (DNC)

3 pre-requisites before conducting DNC assessment

1. Mechanism causing devastating brain injury leading to death
 2. Neuroimaging to support cause
 3. Absence of confounders
 - **Temperature: core ≥ 36 celsius** (rectal, esophageal, bladder, arterial, bladder, central venous)
 - **Time: wait ≥ 48 hrs after arrest** (unless imaging shows devastating injury)
 - **Drugs:** Wait 5 half lives if drug is known (e.g. sedatives, neuromuscular blockers)
 - **Shock:** Must be resuscitated appropriately (i.e. not un-resuscitated)
 - **Metabolic disorders*:** Na 125-159, PO₄ >0.4 , Glucose 3-30, pH 7.28-7.5, PaCO₂ < 60 , Urea < 40 (if available), Cr < 400 , bilirubin < 100
- “If these derangements cannot be corrected and are judged to be potentially contributing to the loss of brain function, ancillary investigation should be considered.”
- Severe weakness: myasthenia, ALS, spinal cord injury
 - Decompressive craniectomy

*Suggestions from Trillium Gift of Life Network (TGLN). Not exhaustive, not necessarily absolute contraindications but require call to donation support physician for advice.

Shemie et al., Can J Anesth (2023) 70:483–557

DNC Assessment

3 criteria to confirm DNC

1. Absence of consciousness – no wakefulness or response to stimuli
2. Absence of brainstem function – no brainstem reflexes
3. Absence of capacity to breathe

1. Motor examination
- Central stimulation (supra-orbital notch)
- Peripheral stimulation

2. Brainstem Reflexes
- Pupillary response to light
- Corneal response
- Cough
- Gag
- Vestibulo-ocular (cold calorics)

3. Apnea testing
- No respiratory effort
- pH < 7.28, pCO₂ > 60 AND
pCO₂ increase ≥20

Oculocephalic / Doll's Eye reflex –no longer recommended
(low specificity / may confound assessment)

Shemie et al., Can J Anesth (2023) 70:483–557

BONUS

Read on own

Apnea Testing

- Pre-oxygenate and obtain ABG (Trillium Gift of Life Network recommends baseline ABG PaCO₂ 35-45, pH = 7.35-7.45)
- Several options for apnea testing
 - CPAP
 - Passive oxygenation (O₂ tubing down ETT, disconnected from ventilator)
 - Exogenous CO₂ or Reduce minute ventilation by 50% if high risk of failing (i.e. from Hypoxaemia)
- Monitor for respiratory efforts
- Serial ABGs at 0, 5, 10 and 15min

Thresholds for completion: PaCO₂ > 60 mmHg AND > 20 mmHg above the pre-apnea baseline AND pH ≤ 7.28.

BONUS

Read on own

DNC: Ancillary Testing

- Required if DNC examination cannot be completed (i.e. confounder cannot be reversed, trauma to one eye prevents pupillary/corneal/vestibulo-ocular reflex testing)
- Ancillary tests used to support theory of cessation of brain function
- Several modalities can be used
 - CT angiography (CTA)
 - CT perfusion (CTP)
 - Transcranial Doppler
 - Radionuclide lipophilic perfusion study
- Don't use: EEG, MRI, 4 vessel angiography, SSEP, BAEP, radionuclide lipophobic perfusion study

BONUS

Read on own

Donation Organ Support [beyond GIM scope]

- Respiratory: Vt 6-8ml/kg, PEEP ≥ 8 , Recruitment manoeuvres
- Cardiovascular: MAP > 65, inotropes (1st Vaso, 2nd Norepi, NO dopamine), crystalloids for fluid resuscitation
- Endocrine: Steroids if on vasopressors, no recommendation for levothyroxine, glucose 6-10
- Diabetes insipidus: Na target 135-155, use vaso or DDAVP
- GI: Continue enteral feeding
- Core temperature 34-35°C if kidneys being considered
- Coronary angiography if heart being considered

Management of Neurological Injuries

Types: traumatic brain injuries, subarachnoid hemorrhages, stroke, hypoxic injuries, etc.

Principles of ICU management (Neuro lecture for specifics):

- Intubation for airway protection (if GCS < 8)
- Head of bed > 45°
- Target normothermia, euglycemia, normocapnia
- Pick normal saline for fluid management: avoid sodium shifts and hyponatremia

Increased ICP management: (in addition to above)

- **Hypertonic 3% saline 250 mL bolus** or **mannitol 0.25–1 g/kg q6–8h**
- **Hyperventilation:** target CO₂ 26–30 mmHg
- **ICP monitoring:** keep < 20 mmHg
- **Optimize sedation** (secure airway first)
- **BP control:** maintain CPP > 60 mmHg
- Avoid tight neck dressings or extreme flexion/rotation
- **Seizure management:** anticonvulsants if suspected
- **Surgical consult:** consider EVD (hydrocephalus) or decompressive craniectomy

Devastating Brain Injury:

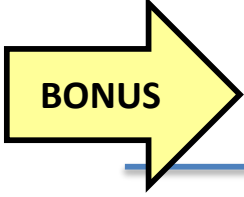
*Incompatible with life or unlikely to make a meaningful recovery.

CCS 2020 recommends waiting ≥72 h before life support withdrawal to improve prognostic accuracy and allow organ donation discussions. Always notify the provincial organ donation program.

Healey et al. Can Journal of ED Medicine 2020; 22(5): 658-660.

Critical Illness Associated Weakness

	Critical illness myopathy	Critical illness polyneuropathy	Glucocorticoid-induced myopathy
Motor	Flaccid quadriparesis, proximal > distal muscles, failure to wean, normal CNs, weak facial muscles	Flaccid quadriparesis, failure to wean, normal CNs	-Gradual onset 1-3 months after starting steroids -Proximal muscle Lower>upper weakness, followed by atrophy of proximal muscle groups -Lower occurs before upper extremity weakness and is more severe
Sensory	Spared	Decreased pin prick/touch in distal extremities	Normal
Reflex	Normal or low	Low	Normal
Other	CK may be elevated Strongly associated with steroid use (starts several days after steroids) -Diagnose with nerve conduction studies/EMGs -No tx	Severe sepsis is the strongest risk factor	-Will get Cushing's like syndrome with , DM, mood alteration, skin fragility, osteoporosis -Definitive dx=decrease steroid dose and see if improves in 3-4 weeks



BONUS

Housekeeping – Guideline Updates

- SCCM had numerous guideline updates in 2023-2025

Clinical Deterioration outside ICU (2024)

- No recommendation for continuous vitals in unselected pts
- Suggest education on recognizing clinical deterioration (conditional, low), include patient/family concerns into hospital early warning systems (conditional, low), & hospital-wide rapid response team (insufficient evidence)

Fever in ICU (2023)

- Measure PO or PR over other methods (weak, very low)
- Avoid routine antipyretic for specific purpose of reducing temp (weak, moderate)
- Suggest PET or CT for fever of unknown etiology (weak, very low), insufficient evidence to recommend WBC scans

Rapid sequence intubation (RSI 2023)

- Incline head/torso (conditional, very low), preoxygenation with HFNC if expecting challenge (conditional, low), or NIV if PFR<150 (conditional low), no difference in induction agents (conditional, moderate), use NMBA when sedative agent given (strong, low)

BONUS

Housekeeping – Guideline Updates

- SCCM had numerous guideline updates in 2023-2024

Stress-Ulcer Prophylaxis (2024)

- Consider critically ill adults at risk for clinically important UGIB (conditional, low/moderate)
- Suggest enteral nutrition (conditional, moderate)
- Suggest prophylaxis (moderate, strong) with PPIs or H2RAS as 1st line (conditional, moderate), at low dose (best practice), either IV or po (conditional, low)

Glucose Management in ICU (2024)

- Treat persistent hyperglycemia (>10) (best practice)
- Suggest not targeting 4.4- 7.7 vs higher 7.8-11.1 due to risk of hypo (conditional, moderate)
- IV insulin $>$ intermittent subq (conditional, very low)
- No statement on CGM or personalized glucose targets (note: some observational data shows benefit in matching target to chronic, pre-hospital level)

Adult Critical Care Ultrasonography CCUS (2024)

- Conditional Recommendation to use to improve clinical outcomes in : Septic Shock, Acute Dyspnea or Respiratory Failure, Targeted Volume management
- Conditional recommendation to use or not use CCUS in adult patients in cardiac arrest.

TOXICOLOGY TIME!

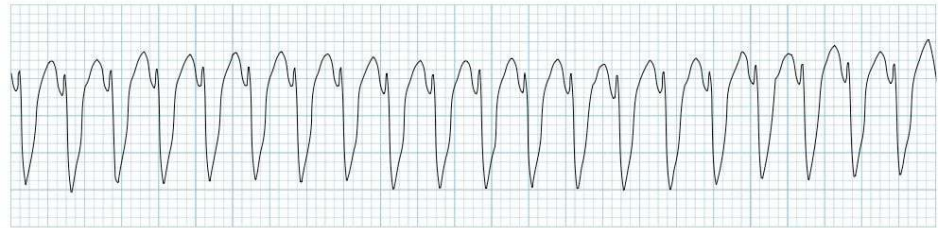
- Toxicology
 - Approach
 - High yield intoxications for studying:
 - TCA
 - Toxic Alcohols
 - Salicylate
 - Acetaminophen
 - Carbon Monoxide, Cyanide
 - Methemoglobinaemia
 - Lithium
 - Hyperthermic Toxidromes
- Extra slides for your reference:
 - Acute hypoxia in the ICU
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MCQ 3 - 2026

A 27M is found down on the steps of his apartment building. Initial vitals are HR 140, BP 120/86, T 38.1, RR 36, SpO2 92% on RA with GCS 10 (E3V3M4). He appears flushed with dilated pupils. There are no medication bottles or drug paraphernalia around him. Upon arrival to the ED, the patient is hooked up to telemetry and the following rhythm is seen. What is the next best step?

- A. Emergent synchronized electric cardioversion
- B. STAT sodium bicarbonate IV push
- C. 1 Amp of D50
- D. STAT 4mg Magnesium IV push



Tox Quick Approach

- History:
 - What did the patient take? How long ago? How much?
 - Co-ingestions (EtOH? ASA? Acetaminophen? Toxic alcohols?)
 - Environmental exposures : Carbon monoxide risk factors
 - Past Medical History, Meds, Allergies, Social
- Physical: look for toxidromes
 - Vitals, GCS, capillary blood glucose
 - Pupils (mydriasis, mioiosis)
 - Neuro (Reflexes, clonus, rigidity)
 - Skin (dry, sweating, hot)



Tox Quick Approach

STAT Investigations

- CBC, Na, K, Cl, bicarb, BUN, Cr, Ca, Mg, PO4, LFTs, CK, troponin, glucose
- ABG, lactate
- Serum osmolality, urea
- **BHCG** (if young female)
- Urine/Serum Tox/**Drug Levels**
 - ALWAYS ask for acetaminophen, salicylates, and EtOH level
- 12-Lead ECG
- Urinalysis (for pH), R+M (for crystals)
- Consider CT head

RC reality:
Labs are usually provided in your question stem, but you will need to calculate AG/OsmGap.

AND SOME QUICK MENTAL CALCULATIONS:

- **Anion gap**= $\text{Na} - \text{Cl} - \text{Bicarb}$ (normal < 12 with normal albumin)
- **Osmolar gap**= $\text{Sesm} - \text{CalcOsm}$ (normal < 10) --> if elevated, order toxic alcohols + start empiric treatment if high suspicion, not explained by ETOH level
- **Calc Osm**= $2\text{Na} + \text{gluc} + \text{BUN}$ ("2 salts and a sugar BUN")

Management

- ABC, IV access, Continuous Cardiac Monitors, O2, Foley
- C-collar if unwitnessed LOC
- Consider – Dextrose, oxygen, naloxone, thiamine
- Tox specific treatment (see next slide)
 - Decrease Absorption
 - Increase Elimination
 - Antidotes
- **Call Poison Control (ALWAYS!)**
 - **And Psychiatry if intentional overdose, once pt is stable and talking**
 - **And Addictions medicine if applicable**

BONUS

Read on own

Gastric lavage not used for decontamination

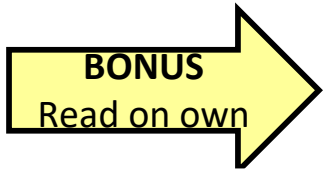
Decontamination/Elimination

Decontamination (prevents absorption of toxin)

Method	Single Dose Activated Charcoal (SDAC) (50-100g)	Whole Bowel Irrigation. E.g. PEG 500ml + 1-2L/hr
Indications	Suitable toxin + 1-2hrs since ingestion >2hrs with drug that delays gastric emptying, massive ingestion, lethal drug	if ingestion large, deadly, long-acting i.e. bupropion, lithium, salicylates
Contraindications	High aspiration risk (reduced LOC, not protecting airway, seizures, vomiting) Poorly absorbed toxins : Metals (Fe, Pb, Li), Alcohols, Salts (K, Na, Mg), Hydrocarbons	High risk aspiration GI: Ileus, obstruction, perforation, bleeding

Elimination (speeds up the removal of absorbed drug already in the system)

Method	Dialysis	Multidose-activated charcoal (12.5g q1h)
Indications	Dialysable drug (small, charged, not protein bound, small volume of distribution) EXTRIP – list of dialysable drugs i.e. Toxic alcohols, ASA, Lithium, Acetaminophen, metformin, CCBs, BBs	Drug cleared by enterohepatic circulation - Phenobarbitol, carbamazepine, theophylline, caffeine, phenytoin, ASA, quinine, dapsone



Common toxins and their antidotes/treatments

Acetaminophen	NAC
ASA	bicarb
BB / CCB	High-Dose Euglycemic Insulin, glucagon, calcium, intralipid
Benzodiazepines	Flumazenil**
Cyanide	Hydroxycobalamin, sodium thiosulfate
Iron	Deferoxamine
Isoniazid	Pyridoxine
Local anesthetics	Intralipid
Lipid soluble drugs (Atenolol, Bupropion, CCBs, Amitriptyline)	Intralipid

BONUS
Read on own

Common **toxins** and their **antidotes/treatments**

Methemoglobinemia

Methylene Blue

Organophosphates

Atropine, pralidoxime

TCA

HCO₃, intralipid (refractory and unstable cases)

Toxic alcohols (ethylene glycos,
methanol)

Fomepizole, thiamine, folate

Valproate

L-carnitine

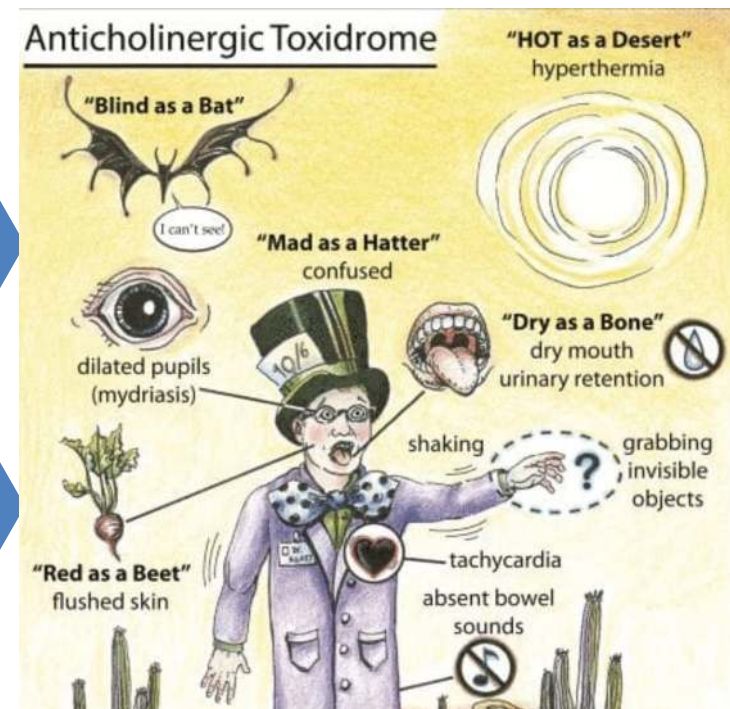
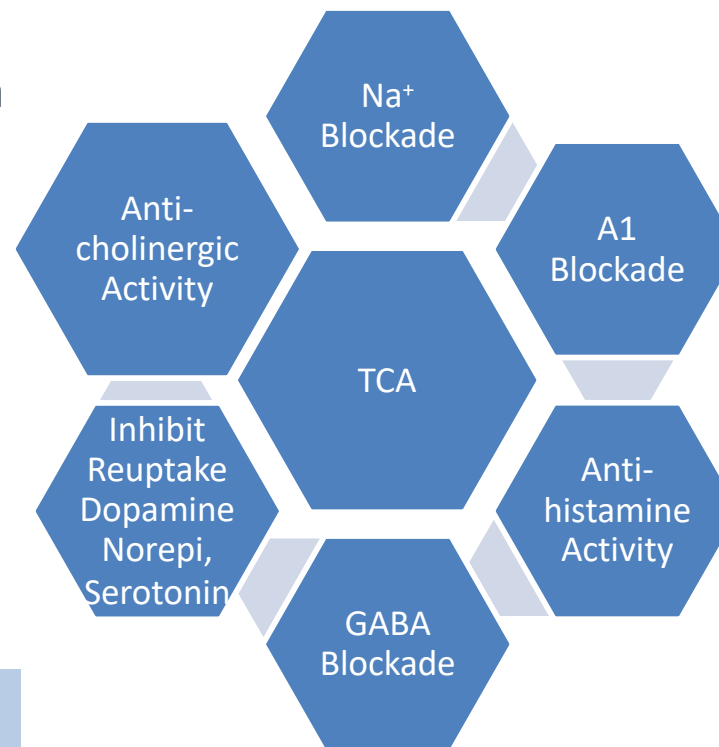
Sulfonylurea

Octreotide

Tricyclic Anti-Depressant (TCA) Overdose

amitriptyline, nortriptyline

- Diagnosis:
 - History of Ingestion
 - Toxidrome
 - ECG Findings
 - Urine tox
- “Dirty Drug” with multiple effects



Beware False + Urine tox:
Benadryl / antihistamines
Quetiapine
Carbamazepine

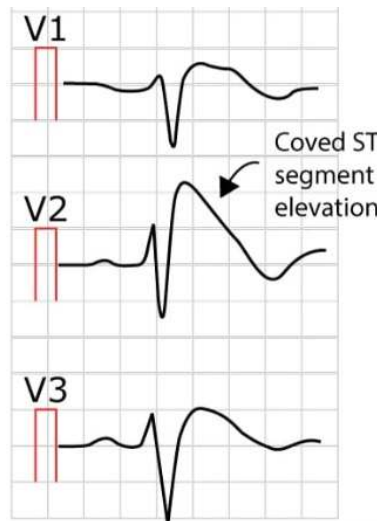


TCA Overdose – Signs & Symptoms



Brugada type I pattern

- CVS
 - Hypotension
 - Arrhythmias
 - Sinus tach
 - VF/VT as QRS widens
- QRS >100
 - >100-26% get seizures
 - >160-50% get arrhythmia



terminal right-axis deviation (RAD) with tall R-aVR



- CNS
 - ↓ LOC, agitation, psychosis, delirium
 - Seizures

TCA Treatment

- ABCs, IV, O2, Monitors, Foley, Acute care area, Poison control
- Decontamination
 - Can consider activated charcoal (1g/kg, max 50g) if present within 1-2 hours unless they have a decreased LOC, gut perforation, bowel obstruction

No role for increased elimination

DO NOT use cholinergic drugs like Physostigmine to counteract anticholinergic toxicity (will worsen cardiac instability, increases risk of cardiac arrest)

TCA Symptom Specific Management

- LOC
 - GCS 8, intubate
 - Agitation: benzodiazepines (Ativan or diazepam at 5 – 10 min intervals)
 - Seizures
 - Start with Ativan or diazepam, add midazolam infusion if refractory
 - Then propofol infusion if refractory
 - Then Barbiturates
- Hypotension
 - NS or Na Bicarb bolus (up to 30ml/kg)
 - Norepinephrine or phenylephrine if refractory
 - Consider hypertonic saline boluses 100 mg IV if refractory despite vasopressors

DO NOT USE PHENYTOIN → Enhances cardiotoxicity as it also blocks sodium channels

TCA - Treatment of Arrhythmias

- Wide complex (ventricular) tachycardia OR prolonged QRS > 100
 - Na Bicarb 1-2mEq/kg IV → if QRS narrows start infusion (3amps in a bag of D5W) at 250ml/h
 - If fails, give magnesium sulfate
 - If fails, lidocaine (class IB) 1.5mg/kg bolus then 1-4mg/min
 - Class IA, 1C, and III anti-arrhythmic are C/I
 - If fail and unstable → lipid emulsion, VA ECMO
- Sodium Bicarb
 - Indication
 - QRS>100, ventricular arrhythmia, or hypotension
 - Goal pH 7.50-7.55
 - Bolus 1-2 amps then run as an infusion at 250ml/h

Toxic Alcohols

	Ethylene Glycol	Methanol
Found in	Antifreeze, wiper fluid, cleaners, fuels, moonshine, solvents, hand sanitizer	
S&S	-Decreased LOC -Frank hematuria, flank pain, oliguria -HypoCa	-Decreased LOC -Retinal injury leading to blindness
PE	-Cranial Nerve palsies -Tetany	-Afferent pupillary defect -Mydriasis (dilation) -Retinal sheen -Hyperemia of the optic disc
Labs	-Classically high AG and OG *see next slide -Lytes, Creatinine and Urine R&M -ECG (Watch QTc for ethylene glycol→ hypoCa→ Prolonged QT)	

What if there is the right history but no Osmolar gap?

- The osmoles get metabolized to anions!
- This means there is no parent alcohol left to inhibit!

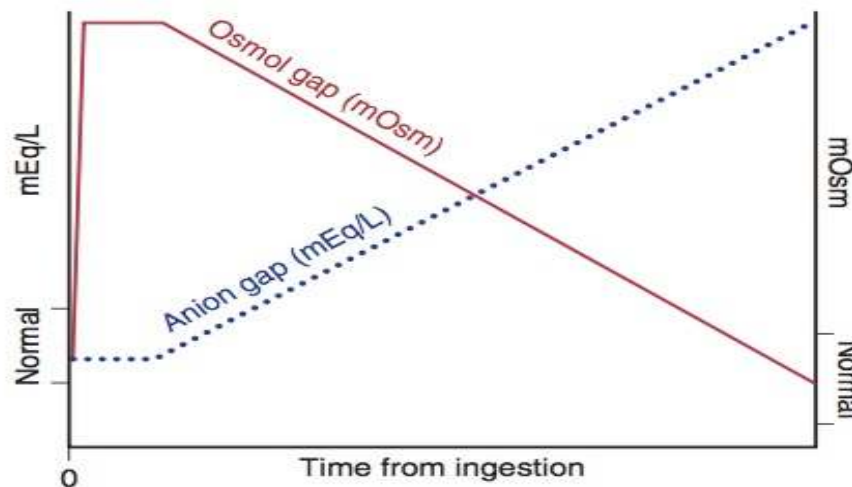


FIGURE 107-4. The reciprocal relationship of anion gap and osmol gap over time (hours). Note that patients presenting early may have a normal anion gap while patients who present late may have a normal osmol gap.

There still may be a toxic alcohol ingestion with a low osmolar gap and high anion gap!

Metabolites:

- Methanol → formate
- Ethylene glycol → Glycolate, glyoxylate, oxalate

The Toxicology Differential of Anion Gap and Osm Gaps

Anion Gap	Osm Gap	Ddx
High	No	Ketones Tylenol (<i>rarely, through 5-oxoproline</i>) Salicylate DKA High lactate (eg shock, status, ischemic gut) Late toxic alcohol ingestion
High	High	Ethylene Glycol Methanol Propylene Glycol ESRD with no IHD
Normal	High	Isopropyl Alcohol Ethanol Severe hyperproteinemia/hyperlipidemia <i>can cause pseudohyponatremia given perceived osm gap.</i>

Treatment

- ABCs, IV, O2, Foley, Cardiac monitor / Monitored setting
- **Decontaminate**
 - No role (can try NG aspirate if within 60min)
- **Enhanced elimination**
 - Acidemia allows toxic metabolites to penetrate end-organ tissue, so give bicarb!
 - Bolus 1-2meq/kg then set up an infusion at 150-250 cc/h
 - Target pH=7.35

Treatment Continued

Alcohol Dehydrogenase Inhibition

- Blocks degradation of parent alcohol / prevents toxic metabolite formation
- **Tx:** Fomepizole or ethanol + cofactors
 - Methanol: Folic acid 50 mg IV q6h
 - Ethylene glycol: Thiamine 100 mg IV + Pyridoxine 50 mg IV

Indications (if high index of suspicion – don't delay treatment!)

- Methanol >6.2 mmol/L OR Ethylene glycol >3.2 mmol/L
or
- Confirmed toxic ingestion of toxic amounts + Osmolar gap >10
or
- Suspected ingestion + ≥ 2 of: pH <7.3 / HCO_3^- <20 / OG >10 / Urine oxalate crystals

Treatment Continued

- Hemodialysis
 - This is the definitive therapy
 - Will clear both the alcohol and their toxic metabolites
 - Indications for HD - any 1 of...
 - End organ dysfunction
(eg. Coma, Seizure, Visual defects, Renal failure)
 - $\text{pH} \leq 7.15$
 - Persistent metabolic acidosis
 - High AG metabolic acidosis
 - Very high level of parent alcohol
- No antidote
- NB treatment of isopropyl alcohol and ethanol is SUPPORTIVE.

Alcohol Withdrawal



- Obtain accurate intake history:
 - Amount, pattern, history of withdrawal
- Diagnosis:
 - Consider wide differential for altered LOC
 - ++ Important as GABAnergic medications can cause harm for other medical disorders
- **Symptoms (typical timeline)**
 - Hallucinoses: 8–12 h
 - Seizures: 12–48 h
 - Delirium tremens: 2–4 d
- **Management – supportive**
 - Symptom Triggered, CIWA-AR most studied
 - BZD most widely used: diazepam preferred as longer acting with active metabolites
 - A word of Phenobarbital - Historically used replaced by BZD without clear inferiority
 - Growing interest among clinicians due to long half-life (3-4 days), linear relationship of dose to drug levels, po/IV/IM all 100% bioavailable
 - 2023 Meta-Analysis: PHB effective in ICU for etoh withdrawal but contradictory evidence of low quality. Further studies needed for dosing, protocol, safety. (Umar et al., CUREUS, 2023)

Salicylate Toxicity

Good Practice Oral
Exam Scenario!

Early Findings (1-2h post-ingestion)	Late Findings
-Tinnitus -Nausea, vomiting -Hyperventilation -Fever	-Coma/seizures -Non-cardiogenic pulmonary edema -Arrhythmia -Thrombocytopenia -AKI

BONUS
Read on own

Salicylate Sources: not just ASA

OTC	Rx	Topical	Herbal
Alka-seltzer (ASA + sodium bicarb)	Aggrenox (ASA+dipyridamole)	Acne cleansers	Red Oil Chinese
Anacin (ASA + caffeine)	ASAcid (mesalamine / 5-ASA = aminosalicylic acid)*	Arthritis topicals	Red / White flower oil
Aspirin (ASA) Ecotrin (also ASA)	Fiorinal (ASA+ butalbital/caffeine)	Oil of wintergreen (methyl salicylate)	Koong Yick Hung Fa Oil
Excedrin (ASA, caffeine, acetaminophen)		Wart removers	Minyak Cap Kapak
Pepto-bismol (bismuth subsalicylate)		Dandruff Shampoos	
Kaopectate (subsalylate)			
Entrophen (ASA)			

BONUS

Read on own

A special word on the brain and Salicylates



- Get neuronal energy depletion as salicylates uncouple neuronal and glial oxidative phosphorylation
- Get a discordance between serum and CSF glucose concentrations
 - Despite normal serum glucose concentrations, CSF glucose concentrations are low in 33%
- May also get cerebral edema resulting in a coma +/- seizures

Salicylates – Investigations

- Salicylate level
 - Toxicity with serum levels greater than 2.9-3.6mmol/L
 - Order levels q 2-4 hours
- ABG/VBG
 - Respiratory alkalosis → direct stimulation of the resp centre
 - Anion gap metabolic acidosis → uncoupling of oxidative phosphorylation
 - Common to see a mixed acid-base disturbance with 2 primary problems
 - If concurrent respiratory acidosis, consider:
 - Acute lung injury
 - CNS depression
 - Mixed overdose

Salicylate Treatment

Decontaminate

- Activated charcoal (1g/kg up to 50g PO/NG) within 2 hours (or longer if enteric coated or bezoar)
- Can consider whole bowel irrigation (ask Poison control)

Enhanced Elimination

- Alkalinize the urine and blood
 - Aim to have a blood pH 7.4-7.5, do not go above 7.55 and urine pH 7.5-8
 - Give 1-2mEq/kg IV bicarb then start an infusion at 250ml/h
 - Watch K, Na and Ca
 - Correct hypokalemia before alkalinization
 - In hypokalemia, the renal tubules will absorb K and excrete H⁺, preventing alkalinization

1. Chyka et al. Clin Toxicol (phila) 2005;43:61
2. Neuvonen et al. Eur J Clin Pharmacol 1978;13:213
3. Goldfrank's Emergency 9th Edition
4. Jurlink et al. Annals of Emergency Medicine 2015

Salicylate Treatment: Enhanced Elimination

- Dialysis indicated if patient has one of the following:
 - Salicylate Levels $>7.2\text{mmol/L}$
 - Hypoxemia requiring supplemental O_2
 - A change in mental status
 - Renal failure (and salicylate level $>6.5\text{mmol/L}$)
 - Progressive deterioration of vital signs
 - Severe acid–base or electrolyte imbalance despite appropriate treatment ($\text{pH} < 7.2$)
 - Hepatic compromise with coagulopathy
 - Volume overload preventing admin of sodium bicarb



Acetaminophen Toxicity

Toxicity:

- Likely to occur with single ingestions > 250 mg/kg or over 12g in 12 hour period
- Severe liver toxicity occurs in doses greater than 350 mg/kg (AST and ALT > 1000) unless appropriately treated

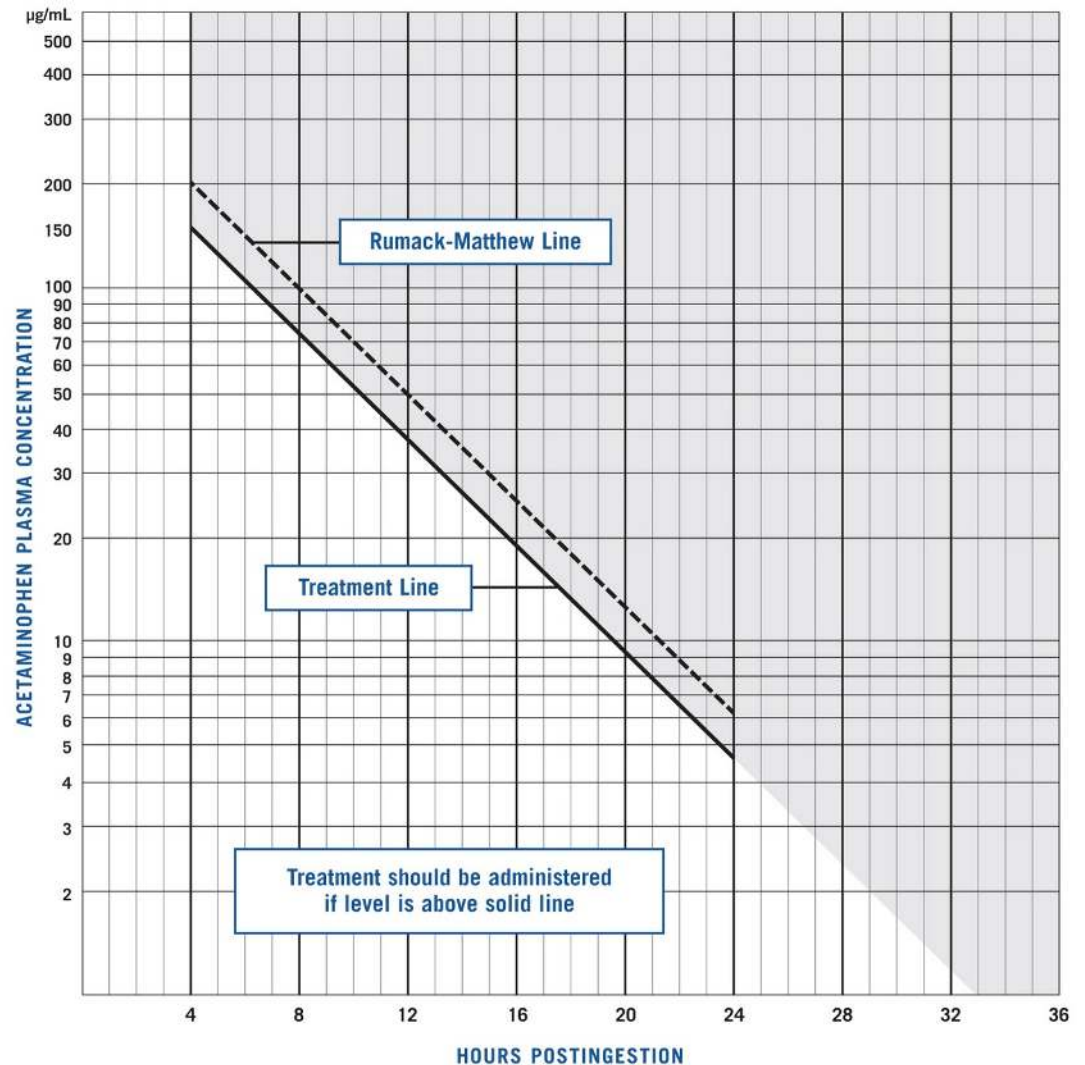
Signs and Symptoms:

- *Stage I (0.5 - 24 hrs)* – nausea, vomiting, diaphoresis, pallor, lethargy, malaise, or asymptomatic
- *Stage II (24-72 hrs)* – RUQ pain, jaundice, oliguria, rise in ALT & AST
- *Stage III (72 – 96 hrs)* – jaundice, confusion (hepatic encephalopathy), marked elevation in AST & ALT, prolonged INR/PTT, bili > 68, hypoglycemia, lactic acidosis, hyperammonemia, bleeding diathesis, renal failure
- *Stage IV (96 hrs – 14 days)* – recovery phase where hepatic necrosis may develop

Treatment

- Acetaminophen levels should be drawn 4 hours after ingestion ideally and plotted on modified Rumack-Matthew nomogram
- Not applicable in repeated supratherapeutic ingestion

Single Acute Acetaminophen Overdose Nomogram



Assessing Risk of Acetaminophen Hepatotoxicity

If ANY of the following occur:

- Ingestion of greater than 7.5 – 10 g in 24 hours
- Ingestion of greater than 4g in 24 hours AND increased risk of susceptibility to hepatotoxicity (ie. Chronic ETOH use)
- Abdominal pain, liver tenderness, nausea, vomiting, jaundice
- Supratherapeutic serum acetaminophen concentrations (> 130 mmol/L)
- Elevated ALT or AST (> 50 U/L) on presentation

Treatment Principles

Gastrointestinal decontamination: activated charcoal (1g/kg or 50g) in patients presenting within 4 hours of potentially toxic ingestion (single dose > 7.5 g)

NAC protocol indications:

- Serum concentration above treatment line on RM nomogram
- Single ingestion > 7.5 g (150 mg/kg)
- Unknown time of ingestion + concentration > 66 $\mu\text{mol/L}$
- History of ingestion + liver injury
- Delayed presentation (>24 h) with evidence of liver injury
- Consider in acute undifferentiated liver failure

Beware NAC infusion reactions.

Mild flushing/urticaria – continue infusion with antihistamine/steroid.

Severe anaphylaxis or angioedema – stop and treat as in allergy online slides

NAC Protocols

20 Hour: 150 mg/kg over 60 min; 50 mg/kg over 4 hrs; 100 mg/kg over 16 hrs.

72 Hour: 140 mg/kg PO loading dose; 70 mg/kg q4h for 17 doses.

Target INR < 1.3, ALT < 100 U/L, acetaminophen < 132) before stopping.

Carbon Monoxide poisoning

Pathophysiology HYPOXIA	Hemoglobin binds CO (COHb) rather than O₂ HbO₂ dissociation curve left shifted =Hemoglobin has more affinity for oxygen, does not release to tissues =Tissue hypoxia despite high SpO₂ CO disrupts oxidative phosphorylation, causes neuronal death
Causes	Carbon combustion + inadequate ventilation (eg) gas stoves Others: Methylene chloride/bromide
Signs / Symptoms	Early (<4hrs): H/A, dizziness, Nausea, 'flu-like' Continued exposure – end organ hypoxia - CNS: Ataxia, syncope, coma - Cardiac: arrhythmias, myocardial ischemia, cardiomyopathy - Resp: pulm edema (cardiogenic or ARDS), dyspnea - Skin: Cherry red skin, skin bullae - Metabolic: AGMA, lactic acidosis - retinal hemorrhages Delayed neurological sequelae [Parkinsonism, dementia, psychosis, apraxia]
Diagnosis	Carboxyhemoglobin (COHb) level: normal 1-2%, smokers 5-10%, >10% abnormal SpO2 will over-estimate - i.e. will read 100% CT/MRI – bilateral opacities in globus pallidus



Carbon Monoxide poisoning

“Confused elderly man comes with headache in winter from remote cabin with normal BP and unexplained AGMA lactic acidosis ...”

Treatment

1. **Remove source, move to well ventilated area**
 2. **100% FiO₂** – HFNC or Intubate (i.e. Coma)
 - COHb half lives
 - Room air: 250-320 minutes
 - 100% FiO₂: 90 mins
 - Increased atmospheric pressure (2.5ATA): 20 mins
 3. **Consider Hyperbaric O₂**
 - Conflicting evidence – **expert consensus**
 - Potential indications
 - If within 6h of exposure is best
 - COHb > 25%
 - Seizure or Coma
 - Syncope
 - Pregnant woman with >15% COHb or Fetal distress in pregnancy
- **If they are suffering CO poisoning after a smoke inhalation, treat for concurrent cyanide poisoning**



Cyanide (CN) Poisoning

Pathophysiology HYPOXIA	CN blocks cytochrome oxidase function Cells cannot use aerobic respiration (toxic to mitochondria) Anaerobic respiration = tissue hypoxia, lactic acidosis
Causes	Inhalation of gases → combustion of plastic, rubber, wool, silk, polyurethane Ingestion: Casaver, apricot pits, bitter almonds Intravenous: Nitroprusside
Signs / Symptoms	3 hallmarks: Coma, Acidosis, Cardiac instability - ABRUPT onset (only delayed to hrs if ingestion) - H/A, Nausea, Dizziness, Confusion - Syncope, Seizure, Coma - Severe lactic acidosis, - Cardiac collapse, cardiac arrest, death - VBG: Venous arterialisation – high venous sat - ABG: High SaO2 (unlike CO toxicity), AGMA - Smell of bitter almonds - ECG: non-specific arrhythmias, blocks, brady, tachy
Diagnosis	Diagnosis is purely clinical – looks for hallmarks and high mixed/central venous sat (>90%) Cyanide levels (NB: in practice Take too long to come back don't wait to treat in emergency): >2.4 coma. >3 death

Cyanide (CN) Poisoning

Treatment

1. Supportive

- Control airway (intubate), 100% FiO₂
- Crystalloids and vasopressors for shock
- Bicarb for acidosis

2. Hydroxycobalamin

- Binds CN forming cyanocobalamin → Urine
- 5g IV over 15mins, repeat x1 if needed
- A/E: anaphylactoid reaction, HTN

3. Nitrites

- MetHb has a higher affinity for CN than Hb → induce methemoglobinemia
- Pre-hospital: amyl nitrate
- Na nitrite 3% 10ml over 2-4mins IV
- A/E: Hypotension, hypoxia (don't give if COHb)

4. Sodium thiosulfate

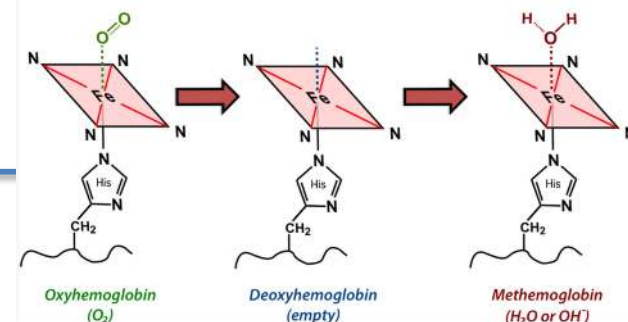
- Enhances renal excretion
- 150-200mg/kg over 10-20mins
- A/E: hypernatremia

1 and 2 are the mainstay of management. 3 and 4 only if hydroxycobalamin is unavailable

Methemoglobinaemia

- Pathophysiology:

- Iron in Hb oxidized to Fe^{3+} by precipitant
- MetHb can't carry O_2 + O_2 dissociation curve L shifted = tissue Hypoxia



Drugs	Local anesthetics (benzocaine, lidocaine) Dapsone Nitrates - Amyl nitrite - Nitric oxide - Nitroprusside - Nitroglycerin Chloroquine Sulfonamides (Sulfamethoxazole)
Hemolysis – any cause	
Genetic	Hemoglobin M NADH methemoglobin reductase deficiency

Clinical Features – based on MetHb level (%)	
3-15	Pulse Ox low (90-95%) Slate gray skin discolouration
15-20	Cyanosis Chocolate brown blood Pulse ox approx 85%
20-50	Dizziness, syncope, fatigue, headache, weakness, dyspnea
>50	CNS depression/Coma Seizures Metabolic acidosis Arrhythmias Death

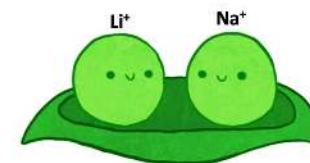
PaO₂ – Saturation Gap

- PaO₂ will be high (pt being treated with O₂)
- SpO₂ reads approx **80-85%**
- PaO₂-SpO₂ mismatch is indicative of hemoglobinopathy

Methemoglobinemia Treatment

- Early recognition
 - Stop offending agent
 - Draw MetHb level, ABG
- Supportive
 - Coma: Airway protection, intubation
 - Cyanosis: Provide high-flow oxygen (if responds then unlikely MetHb)
- **Methylene blue**
 - 1-2mg/kg, repeat at 30 mins if no response
 - Relative Contraindications: G6PD deficiency (may cause hemolysis), serotonin syndrome (serotonergic)
 - No clear cut off for when to use - potentially if symptomatic or MetHb > 30%. Liaise with poison centre.
- Cimetidine – can be used for dapsone induced MetHb
- Ascorbic acid – Theoretically can be used if no response to methylene blue, although not typically used

Lithium Overdose



Pharmacokinetics

- Lithium does not bind to proteins and is distributed freely throughout the body
- Therapeutic window 0.6-1.2mmol/L
 - However, can get sx of overdose in therapeutic window as well!

Signs & Symptoms

- GI-n/v/diarrhea is the earliest symptom CVS-T wave flattening in the precordial leads, QT prolongation, brady, unmask Brugada
- Heme-Increases WBC (15-20)
- CNS (late finding)-Dizzy, lightheaded, orthostatic, lethargy, slurred speech, ataxia, tremor, myoclonic jerks

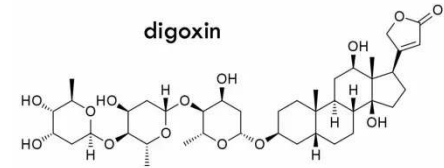
Diagnosis

- Li levels should be obtained on presentation + serially after ingestion of sustained release.

Treatment

- Decontamination-Li is NOT absorbed by activated charcoal. Can try whole bowel irrigation if there is:
 - Sustained release ingestion
 - Symptomatic patients
 - Unknown amounts ingested
 - >40 mg/kg ingested, or <6h from ingestion
 - Increasing lithium levels
- Enhanced elimination-hemodialysis is indicated for patients with:
 - Arrhythmias
 - Seizures or severely abnormal mental status
 - Serum lithium levels $\geq 5\text{mmol/L}$
 - Serum lithium levels $\geq 4\text{mmol/L}$ w/ Cr $>176\mu\text{mol/L}$
- Restore fluid balance (NS at 1.5 maintenance)
- Do NOT enhance elimination with forced diuresis - Get a small increase in elimination but then get salt/water depletion that leads to increased Li retention

Digoxin Toxicity



Mechanism of Action:

- Inhibits Na/K antiporter ($\uparrow \text{Na}^+$ in cell, $\downarrow \text{K}^+$ in cell) $\rightarrow \uparrow \text{Na}^+$ causes $\uparrow \text{Na}^+/\text{Ca}^{2+}$ exchanger leading to more cardiac $\text{Ca}^{2+} \rightarrow \uparrow$ inotropy $\rightarrow$ reflexive $\uparrow$ vagal tone reduces HR

Notable Pharmacokinetics:

- Peak effect 2-6 hours, too large for IHD, half-life 40 hours, renal excretion, many drugs alter excretion (azoles, amio, azithro)

Diagnosis:

- Send level for pts on dig with acute illness. Therapeutic (0.6-2.6 nM/L)
- Chronic >5 nM/L and acute >12.8 nM/L are concerning but levels don't correlate with toxicity

Clinical:

- Cardiac: VARIABLE! Consider in all tachyarrhythmias except rapid atrial fib, sinus bradycardia, heart blocks
- GI: n/v, pain
- CNS: delirium, fatigue, visual disturbances (blurred, halos / "starry night"), rare seizures

Management:

- Decontaminate: Activated Charcoal within 1 hour
- Antidote: **dig specific antibody fragments (Digibind, Digifab)**. Definitely give if dysthymias, hemodynamically unstable, $\text{K} > 5$ (secondary to dig). Maybe give if severe GI sx, >10 mg ingestion, renal failure, ?altered LOC
 - "CALL POISON CONTROL" to confirm if giving
- Correct Mg^{2+} , correct K^+ (low K^+ = good, high K^+ poor prognosis), avoid Ca^{2+}
- Note that dig half-life $<$ FAB half-life, watch for rebound

Organophosphate Poisoning

- Typically found in insecticides
 - Also have applications in medical uses such as reversing neuromuscular blockade (neostigmine/physostigmine), or treating myasthenia gravis and alzheimer's (pyridostigmine, donepezil, edrophonium)
- Mechanism of Action
 - Bind acetylcholinesterase and render it nonfunctioning
 - This will lead to an overabundance of acetylcholine at the neuromuscular junction
- S&S
 - Onset within 3 hours if oral or resp exposure
 - Dermal may be up to 12h
- Muscarinic effects: DUMBELS
 - Diaphoresis, Diarrhea
 - Urination
 - Miotic pupils (SMALL)
 - Bronchospasm, bradycardia, bronchorrhea
 - Emesis
 - Lacrimation
 - Salivation
- Nicotinic effects: MATCH
 - Muscle weakness (paralysis)/fasciculations
 - Adrenergic stimulation...mydriasis (large pupils)
 - Tachycardia
 - CNS-lethargy, seizures, coma, resp depression
 - HTN
- CV: QTc prolongation, MI's, CV collapse (? Secondary to vasodilation)
- Resp: Resp failure secondary to CNS depression, neuromuscular weakness (diaphragmatic weakness), resp secretions, bronchoconstriction
- Neuro:
 - Nicotinic effects
 - "intermediate syndrome"
 - 24-96h after exposure
 - Neck flexion, decreased reflexes, CN abnormalities, prox muscle weakness and resp insufficiency
 - Organophosphate agent induced delayed neuropathy
 - 1-3 weeks after ingestion
 - Painful stocking and glove paresthesia followed by a symmetrical motor polyneuropathy characterized by flaccid weakness of the lower extremities

Dx

- Based off history

Treatment

A/B

- 100% FiO₂
- Intubate (constant reassessment of ability to protect airways and resp insufficiency from diaphragmatic weakness)
- Avoid succinylcholine as it is cleared by acetylcholinesterase
- Can use rocuronium, but will need high doses

C-Usually hypotension and bradycardic

- Can cause HTN and Tachy with sympathetic stimulation

D-Decontaminate-Wash patient, medical personnel should wear full protection, be in well vented area

Antidote

- **Atropine**

- Atropine competes for acetylcholine, preventing cholinergic activation
 - Fixes muscarinic symptoms
- Indications:
 - Miosis
 - Excessive sweating
 - Hypotension
 - Resp distress (poor air entry, bronchorrhea, bronchospasm/wheeze)
 - Bradycardia
- Start at doses 2mg IV, double dose q3-5 minutes until pulmonary signs and symptoms are alleviated
 - Once atropinization, start infusion 10-20% of total dose required to atropinize patient per hour (to max 3g/h)
- S/E
 - Agitation, urinary retention, ileus, hyperthermia, tachy causing MI if pre-existing heart disease
 - Tachy is NOT a C/I to cholinergic toxicity-may be tachy from hypoxia...which will resolve!

BONUS

Read on own

Summary of Common Toxidromes

Toxin	Signs & Symptoms	Treatment
Anticholinergic (TCAs, antihistamines, sleep aids, atropine, etc.)	aLOC (delirium, seizures, coma) Tachycardia Mydriasis, hyperthermia, dry membranes Urinary retention	ABCs Generous IV fluids PRN Prolonged QRS -> Na HCO ₃ Seizures -> benzos Consider physostigmine (ask poison control)
Cholinergic (organophosphates, insecticides, farm products, nerve agents, neostigmine, donepezil)	aLOC Bradycardia, hypotension Emesis, diarrhea, salivation, bronchorrhea Miosis	ABCs Generous IV fluids PRN Supportive care Severe cases often require intubation Seizures/agitation -> benzos Consider atropine (ask poison control)
Salicylate (ASA, Pepto Bismol, OTC oils/creams)	Characteristic Tachycardia, hyperthermia & hyperpnea Mixed met acidosis/resp alkalosis classic Tinnitus, vomiting, aLOC	ABCs, supportive care Generous IV fluids PRN Glucose NaHCO ₃ Consider dialysis

NB investigations are pretty much the same for all (CMP, ECG, osmolality, bHCG, VBG, etc...)

BONUS

Read on own

Summary of Common Toxidromes 2

Toxin	Signs & Symptoms	Treatment
Toxic alcohols (methanol, ethylene glycol)	aLOC EG – hematuria, AKI, flank pain Methanol – Visual symptoms, blindness Elevated osmol gap, later anion gap	ABCs, supportive care Consider fomepizole Folic acid, thiamine and pyridoxine Consider hemodialysis Isopropyl alcohol – supportive care ONLY
Acetaminophen	Early (0-24h): malaise, nausea/vomiting, abdominal pain Later (24h+): RUQ pain, jaundice, aLOC, progressive liver failure - Elevated AST/ALT, functional liver impairment	ABCs NAC as per Rumack-Matthew Nomogram Supportive care
Sympathomimetic (Amphetamines, cocaine, MDMA)	Tachycardia, hyperthermia, hypertension Mydriasis, diaphoresis aLOC, agitation, seizure, coma Cardiac events, elevated CK, acute renal/liver failure, neurologic events	ABCs, supportive care (cooling, sedation) Benzos first line for agitation, HTN, TK If cocaine, consider ASA for cardiac ischemia and nitro Also consider phentolamine +/- CCBs

NB investigations are pretty much the same for all (CMP, ECG, osmolality, bHCG, VBG, etc...)

BONUS
Read on own

Summary of Common Toxidromes 3

	Methemoglobinemia	Carbon Monoxide	Cyanide Toxicity
Scenario	Dizzy/confused after exposure to dapsone/Septra/nitrates	Confused man w headache from his cabin in the winter, AGMA Lactic acidosis	“Caught in industrial fire, now looks very sick, coughing but sat is high”
Causes	Many drugs – nitrates, sulfa...	Carbon combustion w/ poor ventilation	Combustion of plastic/wool/etc Toxic ingest (bitter almond, apricot pits)
Signs/ Symptoms	Cyanosis Chocolate brown blood	Cherry red skin Headache, dizzy, “flu like” Cardiac arrhythmias	Coma, Acidosis, Cardiac instability CNS symptom, sz, coma Severe lactic acidosis AGMA
Diagnosis	ABG, MetHb SpO2 80-85 w high PaO2 “PaO2-SpO2 mismatch” suggests hemoglobinopathy	SpO2 high AGMA with high lactate Carboxy-Hb level (>10% abnormal, Normal 1-2%, smokers 5-10%)	VBG high venous sat ABG high SaO2, AGMA and severe lactic acidosis
Antidotes	Methylene Blue 1-2mg/kg	100% FiO2	Hydroxycobalamin 5g IV over 15. min
Other Rx	Supportive	Hyperbaric oxygen rarely (for severe cases eg seizure/coma or pregnancy related complications)	Sodium thiosulfate (enhances renal excretion) Nitrites (amyl nitrate/sodium nitrite) – induce MetHb which has higher affinity for CN than regular Hbg

After the Opioid overdose...

Required Recovery Reading

Medication	Mechanism	Prescribing Notes
Naltrexone Use for both: Opioid Use Disorder Alcohol Use Disorder	Competitive antagonist of Mu- receptor Usual dose 50mg per day	With Opioid Use Disorder this is not first line, consider if pt unable/unwilling to use others below. Can improve retention in program and reduce incarceration. Must be OFF opioids 7-10d to avoid precipitated withdrawal. Can reduce frequency and amount of ETOH
Buprenorphine/ Naloxone* (SUBOXONE) or IM Buprenorphine (SUBLOCADE) q28d For Opioid Use Disorder	High affinity partial agonist at mu-receptor; at high doses saturates receptor blocking effect of other opioids. *Naloxone orally = ++hepatic first pass effect, so no significant systemic effect. BUT if Suboxone is INJECTED however naloxone component will ppt withdrawal.	Pt must be in withdrawal 12-24h opioid free to initiate. Standardized protocols using the Clinical Opioid Withdrawal Scale (COWS) to titrate dosing exist.
Methadone For Opioid Use Disorder	L-enantiomer binds mu-receptor, high intrinsic activity ~ opioid, low affinity; S-enantiomer acts as NMDA antagonist	Can start immediately Need special license to prescribe **QT prolonging**

+ Social Work, Addictions Medicine referral
 + Harm Reduction: Naloxone kits, safe injection practices/sites

BONUS

Read on own

Toxic Hyperthermia

	Serotonin Syndrome	NMS	Malignant Hyperthermia	Heat Stroke (new guideline SCCM 2025)
S&S	- Autonomic (tachy, hypertension, vomit, fever, diarrhea, diaphoretic) - Neuromuscular hyperactivity (tremor, muscle rigidity lower>upper, myoclonus, hyperreflexia, bilateral babinskis, ocular clonus) - Change in mental status (anxiety, agitated, restless, disoriented)	Tetrad "FARM" - Fever-Temp>38, may be >40 - Autonomic-Tachy, labile BP, arrhythmia, diaphoresis - Rigidity-Lead pipe/cogwheel increase tone NO CLONUS –They are HYPOREFLEXIVE - Change in mental status - Agitated delirium, Catatonia, coma	Rare genetic (autosomal dominant) condition VERY HIGH temp (>42oC often) with diaphoresis Extreme Rigidity Hyporeflexia	- CNS (headache, agitation, confusion, coma, seizure) - Tachycardia, high temperature, +/- tachypnea, +/-hypotension - Weakness, nausea/vomiting, profuse sweating OR dry, signs of dehydration +/- Rhabdo, liver abnormalities, AKI
Onset	Within 24h	Days to weeks	Time from drug - 30min – 24h	Post exposure
Off	24h	2 weeks		Hours

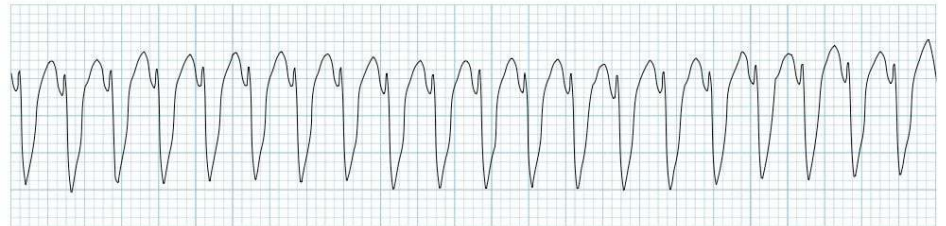
	Serotonin Syndrome	Neuroleptic Malignant Syndrome	Malignant Hyperthermia	Heat Stroke
Inciting Meds	-SSRIs -Amphetamines, Cocaine, MDMA, Levodopa -TCA, SNRI, NDRI -Tramadol, Meperidine -St John's wart, VPA -MAOI -Direct serotonin agonists→Triptans, Ergot, Fentanyl, buspirone	-All classes of neuroleptic drugs can be implicated (Quetiapine, clozapine, risperidone, olanzapine, etc) -Antiemetic agents (domperidone, metoclopramide, prochlorperazine) Acute withdrawal of L-dopa therapy in pt w severe Parkinson's has also been reported to trigger NMS	Reaction to inhaled volatile anesthetics and depolarizing NM blockers (succinylcholine) Can occur after 1st exposure but more likely after 3rd exposure	Environmental exposures No drugs "cause" heat stroke Elderly patients taking sedating meds/SSRIs may be more susceptible. Pay attention to those taking diuretics, antihistamines, antidepressants
Diagnostic criteria	Hunter Criteria: Needs to take a serotonergic agent and ONE of - Spontaneous clonus - Ocular clonus - Inducible clonus + diaphoresis or agitation - Tremor + Hyperreflexia - Hypertonic + temp>38 PLUS ocular or inducible clonus	No criteria	Anesthesia will notice this intra=op – rise in ETCO₂, rigidity, Temp >1oC higher Genetic tests (RYR1 gene most commonly or muscle biopsy – beyond scope of Gim exam)	1. Core temp > 40C 2. CNS abnormalities 3. Non exertional (environmental) vs exertional Elderly, athletes, infants at particularly high risk
Treatment	-Stop the agent, support -Sedate with benzos (goal is to eliminate agitation, hypertonia, normalize vitals) -If fails→cyproheptadine	-Stop the agent, Support, Cooling blankets -Benzos are mainstay -Dantrolene and Bromocriptine <u>are adjuncts</u>	<u>DANTROLENE</u> And rapid cooling , supportive measures	-ABCs -Continuous temp monitoring -Rapid cooling (bear hugger, ice packs, IV cooled saline, bladder irrigation) to <~38-39 C - DO NOT use Dantrolene

MCQ 3 - 2026

A 27M is found down on the steps of his apartment building. Initial vitals are HR 140, BP 120/86, T 38.1, RR 36. SpO2 92% on RA with GCS 10 (E3V3M4). He appears flushed with medication bottles or drug paraphernalia in the ED, the patient is hooked up to a monitor and a rhythm is seen. What is the next best step?

- A. Emergent synchronized electric cardioversion
- B. STAT sodium bicarbonate IV push**
- C. 1 Amp of D50
- D. STAT 4mg Magnesium IV push

B is correct. The overall toxidrome here is concerning for a TCA overdose. Ventricular tachycardia is a rare but potentially fatal complication of this. In the context of a toxidrome, the next best step would be to push bicarbonate prior to attempting an electrical cardioversion.



Thank You!

- Shock and Sepsis
 - Respiratory failure & ARDS
 - Extubation
 - ICU Delirium, Sedation, Analgesia, Sleep
 - Targeted temperature management, neuroprognostication, NDD, neuro ICU
 - Toxicology
- Extra slides for your reference:
 - AIR EMBOLISM
 - Acute hypoxia in the ICU
 - Gas trapping
 - Maternal cardiac arrest
 - Hyperthermia & Hypothermia
 - Bonus MCQs

BONUS

Read on own

Air Embolism

Definition

- Venous air embolism: air -> veins -> RV -> pulmonary circulation
- Arterial air embolism: air -> artery -> end-organ

Etiology

- Surgical (esp NeuroSx, ENT)
- Trauma
- Iatrogenic (e.g. central venous catheters)
- MISC (Barotrauma, pregnancy, ECMO...)

Signs & Symptoms

- Sudden respiratory collapse
- Sudden neurologic decline

Signs & Symptoms (contd.)

- Tachycardic, tachypnea, hypoxemia +/- hypotension, cardiac arrest
- End-organ failure (arterial)
 - Basically any end-organ may be affected depending on location of embolism (think arterial clot)
 - Brain/stroke most common
- Sinus TK +/- RH strain, low plat, elevated CK, pulmonary edema
- Imaging may be NORMAL (air gets absorbed)

Investigations

- CMP, ECG, TTE, contrast CT brain/chest/AP

Treatment

- ABCs, supportive care (pressors PRN, 100% FiO2)
- Position left lateral decubitus if venous, supine if arterial
- Hyperbaric oxygen

BONUS

Read on own

Acute Desaturation/Hypoxia

1. Check the ventilator – all connections intact? Is the oxygen connected?
2. Disconnect ETT from vent
 - Bag-ventilate – is there increased resistance?
 - AIRWAY = blocked ETT, bronchoconstriction
 - Airspace = Blood, Pus, Water, Cells, Protein
 - Pleura = Pneumothorax, effusion, hemothorax
 - Vascular = Pulmonary Embolism
3. Deep suction
4. Auscultate, check that trachea is midline**
5. Check other vitals – hypotension, tachycardia = could it be hypoperfusion?
6. Chest X-ray → check ETT placement (right mainstem intubation?)
7. Review history: New Line? (pneumo)] New blood? (TRALI/TACO) ACS/bolus? (acute pulmonary edema) Off DVT proph? (PE) Fevers? (VAP)

BONUS

Read on own

Acute Respiratory Distress in the ICU

Auscultate and Check that Trachea is midline:

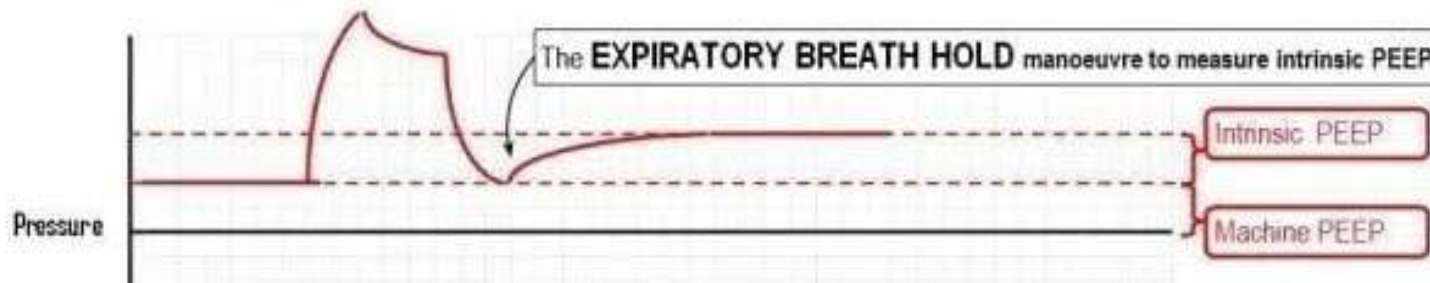
Diagnosis	Trachea Displacement	Air Entry	Percussion	Other
ETT migrated (commonly to R mainstem bronchus)	To LEFT (if R mainstem intubation)	Decreased on L	Decreased on Left	
Pneumothorax	Away from affected lung	Decreased on affected side	Increased on affected side	May have SubQ emphysema if trauma induced
Collapse (mucous plug)	Towards affected side	Decreased on affected side	Decreased on affected side	

BONUS

Read on own

Gas Trapping

- Aka auto-peep or intrinsic-peep
- What is it
 - Occurs when the expiratory time on the vent is shorter than the actual time needed to fully deflate the lungs
- How to detect
 - Do an end-expiratory breath hold



BONUS

Read on own

Gas Trapping

Signs and Symptoms

- Increased work of breathing
- Wheeze
- Increased chest distension
- Decreased chest expansion
- Bilateral decreased air entry
- Increased CO₂
- Increased intrathoracic pressures
 - Dec venous return and hemodynamic instability

Causes

- Machine factors
 - Kinked ETT
 - ETT clogged by sputum
 - Patient biting on ETT
- Vent settings
 - High RR
 - High I:E ratio
- Patient
 - Bronchospasm
 - Increased RR

BONUS

Read on own

Gas Trapping

Patient changed

- Reverse anything reversible (bronchodilators/steroids)
- Suction ETT and make sure patent

Vent changes

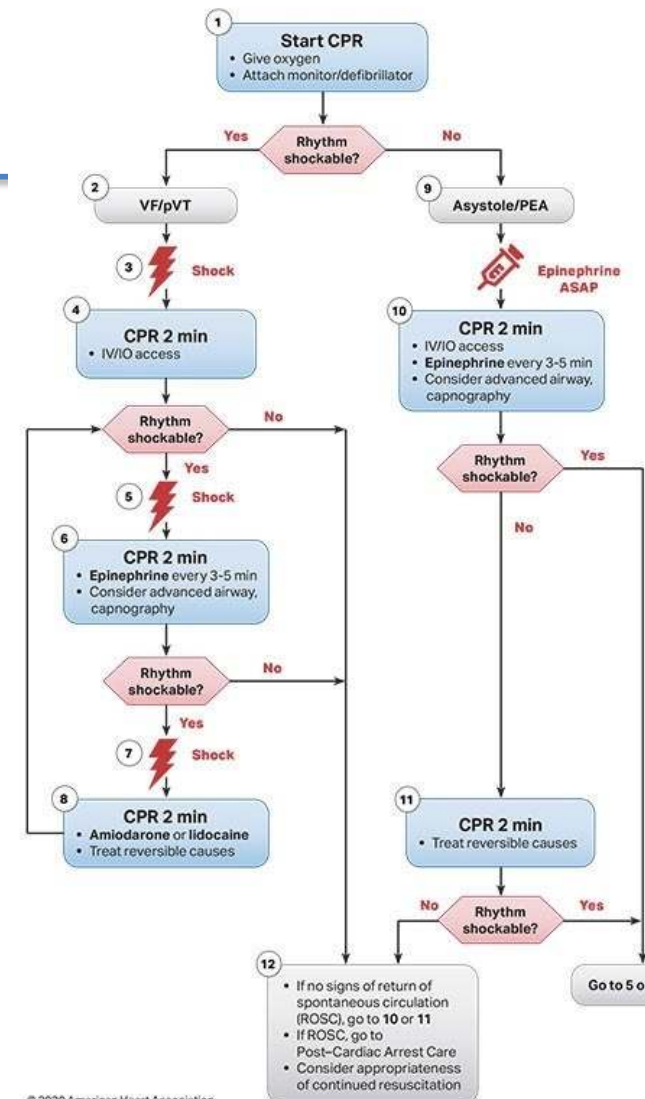
- Need long I:E ratio (1:4 or 1:5) (i.e. give more time to exhale)
- Lower the resp rate
- Decrease the Vt
- Apply PEEP to counter the increased work of breathing
- Last line measures
 - Disconnect from vent and press on chest
 - Heliox
 - ECCOR2
 - High frequency oscillation

BONUS
Read on own

Maternal cardiac arrest

- Run ACLS as one would non-pregnant patient
- Detach all fetal monitors
- If it is a shockable rhythm-defibrillate!
 - A delay in defibrillation decreases chance of survival
- Must move the gravid uterus off the IVC once fundus is at or above umbilicus (20 weeks)
 - IVC compression reduced SV and CO → **Manual** left lateral uterine displacement
 - IV must be above the diaphragm
 - If getting IV Mg, stop and give calcium chloride or gluconate
- Focus on early intubation as ++hypoxia (either ETT or supraglottic device)
- Resuscitation should not be overshadowed by postmortem cesarean delivery
 - PMCD should be considered in the later half of pregnancy
 - Consider PMCD at 5 minutes of resuscitation

Adult Cardiac Arrest Algorithm (VF/pVT/Asystole/PEA)

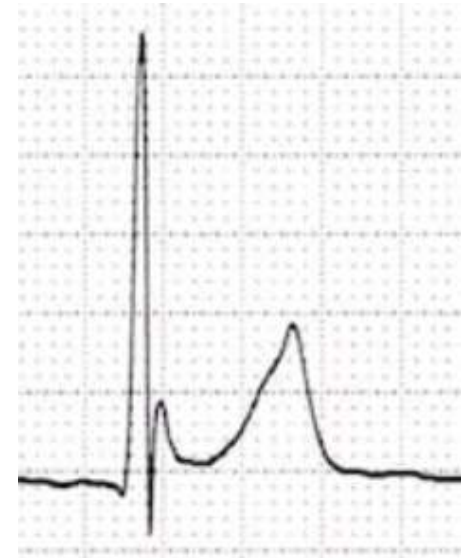


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Hypothermia

- Clinical Effects: bleeding, immune suppression, arrhythmias
 - ECG: **Osborne (J Waves)**, Bradyarrhythmia, PR/QT/QTC prolongation
- Identify Cause:
 - Obvious Triggers: Cold exposure, surgery
 - Distributive Shock
 - Metabolic: hypoglycemia, hypothyroid, AI, DKA, thiamine deficiency
 - Neurologic: thalamic stroke, SCI
 - Toxicologic: anything that can severely sedate



BONUS

Read on own

Hypothermia

Staging & Management				Rewarming Techniques	
Stage	Clinical	Core Temp	Treatment	Technique	Rate (°C/hr)
HT1	Conscious, shivering	35-32	Warm environment/clothes, warm sweet drinks, active movement	Warm room/clothing/drink + movement	2
HT2	aLOC, no shivering	<32-28	Cardiac monitors and limit movement Full body insulation, active external + minimally invasive rewarming (forced air, warm IVF)	As above + Minimally Invasive (forced air, warm IVF)	0.1-3.4
				Peritoneal Dialysis	1-3
				IHD	2-4
				Thoracic Lavage	3
HT3	Unconscious, not shivering, vital signs present	<28-24	As above + airway prn. Consider ECMO if unable to med manage arrhythmias	VV ECMO	4
HT4	No vital signs	<24	As above + CPR, epi x3, defib	VA ECMO	6
				Cardiopulm Bypass	9

Brown D et al., N Engl J Med 2012; 367:1930-1938

BONUS

Read on own

Critical Care Evidence Update

Van Baarle NEJM 2023

- safe to not give proph plts before CVC
- Plt count of 10-50 in ICU and Heme Ward
- Withholding transfusion non inferior

APROCCHSS Trial -BOSCH et al, JAMA IM 2023

- Fludrocort + Hydrocort vs Hydrocort alone for septic shock
- Retrospective observational cohort
- Additional of fludrocortisone was superior to hydrocortisone alone across multiple outcomes including mortality

SCCM 2024 Clinical Update

- Recommend Corticosteroids for severe CAP (strong recommendation, moderate certainty)
- Severe = PNA severity index >4 , CURB-65 >3 , admit to ICU, vasopressors, multi-lobar PNA on CXR
- Various dosing, CAPE-COD trial (NEJM 2023) Hydrocort 200 IV/d x 4 d vs 8 days depending on clinical status then taper over 8 to 14 days

BONUS

Read on own

Bonus MCQ#1 2023

A 25 year old female presents with an out of hospital cardiac arrest due to a cocaine overdose. After a prolonged resuscitation by EMS, return of spontaneous circulation is achieved after 40 minutes. The patient is then admitted to ICU for further management. The patient does not respond to painful stimuli, has an absent pupillary response and absent corneal reflex. There is no cough or gag reflex detectable. You are concerned about potential brain death. What prevents you from making this diagnosis?

- a) Norepinephrine required to maintain a MAP of 65
- b) Cardiac arrest occurred less than 24 hours ago
- c) Serum sodium of 158
- d) Propofol infusion that stopped only 12 hours ago

BONUS

Read on own

MCQ #2 (2023)

A 45 year old male with morbid obesity is admitted for respiratory failure secondary to pneumonia. He has a prolonged course on the ventilator and underwent tracheostomy 5 days ago. He becomes agitated and today he pulls out his trach.

What do you do?

- a) Reinsert the trach
- b) Orotracheal intubation
- c) Cricothyroidotomy
- d) Place on nasal prongs and monitor

BONUS

Read on own

MCQ #3 (2023)

A 25 year old female presents with an acute Tylenol overdose. She ingested 8.5 g four hours before presentation. She is asymptomatic currently, and was given activated charcoal. Her GCS is 15 and she is vitally stable. Her exam is remarkable for mild right upper quadrant tenderness. Her labs are unremarkable, AST and ALT are within normal limits. Her Tylenol level is above the treatment line on the RM nomogram. She is started on a NAC infusion. Thirty minutes into the infusion she develops urticaria and flushing. The infusion is stopped, she is given benadryl and the reaction subsides.

What is the best course of action?

- a) Do not restart the infusion and consult Nephro for hemodialysis
- b) Switch to PO NAC
- c) Reduce the dose by half and uptitrate if the infusion is tolerated
- d) Restart the infusion and monitor for airway compromise or signs of anaphylaxis

BONUS

Read on own

MCQ #4 (2023)

A 66 year old obese male is admitted to the ICU with acute hypercarbic respiratory failure secondary to a severe COPD exacerbation. He presents with a CO₂ of 85, and with this has a GCS of 10. He is started on steroids, routine puffers, and trialed on BiPAP, and after 2 hours his GCS is 8 and his CO₂ has climbed to 105. The decision is made to intubate the patient. His initial vent settings are: ACPC 14/5, 40% FiO₂, RR 18. Peak pressures are 35 cm H₂O, plateau pressures are 20 cm H₂O, driving pressure is 15 cm H₂O, and tidal volume is 8 ml/kg. You are called to his bedside 3 hours later because the patient has acutely desaturated, and is requiring an FiO₂ of 80%. His SpO₂ is now 85%. How would you advise adjusting the ventilator settings to improve his oxygenation?

- a) Increase the FiO₂ to 100%
- b) Increase the respiratory rate to 25 breaths per minute
- c) Switch to volume control ventilation
- d) Increase the PEEP to 10 cm H₂O

MCQ 5 (2024)

65 year old patient is admitted to the ICU after an unwitnessed out-of-hospital cardiac arrest secondary to fentanyl overdose. It is now 60hrs since the arrest. CT head shows diffuse grey-white matter loss in keeping with a hypoxemic-ischemic brain injury, with cerebral edema and tonsillar herniation. Post-cardiac arrest management included therapeutic hypothermia targeting 35-36 degrees C. Passive rewarming has started and the current core temperature is 35.6 degrees C. The patient was sedated with propofol and ketamine which were stopped 24hrs after the cardiac arrest. No neuromuscular blockers were used. MAP is currently 70 with vasopressin, ABG showing pH 7.33, PaCO₂ 45, lactate 1.7. Bicarb 24. Further labs show Na 150, K 3.4, Cr 140, Urea 15, Calcium 2.1, Mg 0.8, Phosphate 0.7, Bilirubin 20, ALT 60, ALP 80.

Which of the following would prevent death by neurological criteria assessment?

- a) Core temperature
- b) Time since cardiac arrest
- c) Metabolic derangement
- d) Shock
- e) Sedative medication

MCQ 6 (2024)

80 year old patient presents to the emergency department with nausea and vomiting secondary to a small bowel obstruction. Past medical history includes chronic pain, depression, cholecystectomy. Home medications are citalopram, hydromorphone contin, senna. The patient undergoes an urgent laparotomy with resection of adhesions for management of the small bowel obstruction. They are admitted to the ICU for post-operative management. Issues during the ICU stay include post-operative ileus, delirium and pain. ICU medications include hydromorphone CR, IV fentanyl, IV gravol, IV metoclopramide, lansoprazole, citalopram, dalteparin 5000 u sc daily. On post-operative day 3 the patient develops a temperature of 39 degrees C. Vitals shows BP 130/80, HR 110, RR 20, Sats 96% 2L NP. Examination shows pupils 5mm and equal, diaphoresis, agitation, lower limb rigidity with brisk reflexes, ankle clonus x 12 beats bilaterally.

Which syndrome most likely explains the findings?

1. Sympathomimetic toxidrome
2. Anti-cholinergic toxidrome
3. Serotonin syndrome
4. Neuroleptic malignant syndrome
5. Malignant hyperthermia

MCQ 7 (2024)

A patient presents 4 hours after an ingestion of an unknown quantity of pills and clear liquid. They are initially confused and agitated, and subsequently become drowsy. Bloodwork shows Na 140, K 4.5, Cl 105, bicarb 21, urea 10, glucose 5, VBG pH 7.30 pco2 40, lactate 2.4, serum etoh 30 mmol/l, plasma osmolality 395, serum/urine ketones positive, acetaminophen 80 micromol/l (upper limit normal < 66, rumack-matthew treatment threshold >800 micromol/l at 6hrs)

What is the next best step in management?

1. D10 infusion
2. NAC infusion
3. Dialysis
4. Fomepizole
5. Single Dose Activated Charcoal

MCQ 8 2024

A 50 year old female presents to the emergency department after starting a new antibiotic for a UTI. She has a headache, dizziness, shortness of breath and confusion. On assessment vitals show BP 110/70, HR 110, RR 28, Sat 87%, temp 36.5. She appears cyanotic, tachypneic and drowsy. She is placed on a non-rebreather mask at 15L/min. Laboratory investigations show Hb 115, WBC 8, Plt 250, Cr 120, Na 137, K 4.8, Cl 105, bicarb 21, lactate 2.3, ABG 7.44, CO2 35, PaO2 170, SaO2 56%.

What is the definitive next step in management?

- a. Hydroxycobalamin 5g IV
- b. Place on high-flow nasal cannula at FiO2 1.0
- c. Methylene blue 2mg/kg IV
- d. CT head
- e. Transfuse 1 unit RBCs

MCQ #9 –2023

A 56 year old male presents with ARDS secondary to acute smoke inhalation during a condominium fire. He is intubated on arrival to the ED and transferred to the ICU. He is diagnosed with ARDS and his ABG PaO₂ on day 3 is 87, despite deep sedation, INO and paralysis. His vent settings are as follows: ACVC Vt 4ml/kg, PEEP 14, FiO₂ 100%, pPlat 30, driving pressure 14.

- What is the next best step?
 - a) Give methylprednisolone 125 mg daily
 - b) Reduce PEEP to 10 and repeat an ABG
 - c) Diurese with Lasix 40 mg IV BID
 - d) Place patient in prone positioning for 16 hrs



MCQ 10 -2024

75 year old patient is admitted to the ICU after an out-of-hospital cardiac arrest. The patient was found by bystanders. No CPR was initiated before paramedic arrival. The initial rhythm was ventricular fibrillation, they required multiple shocks for Vfib/VT, with a total downtime of approximately 20 minutes.

In the ER: Intubated and diagnosed with a STEMI. They underwent PCI with angioplasty to the LAD. Post-cardiac arrest management included therapeutic hypothermia targeting 35-36 degrees C. It is now 96 hrs since the arrest and the following information is available:

Examination shows absent pupillary reflexes bilaterally, absent corneal reflexes bilaterally, M1 on motor examination. CT head shows subtle reduced grey-white matter differentiation with a ratio of 1.5.

When considering neuroprognostication for this patient, which of the following results best predicts a poor neurological outcome (cerebroperformance categories 3-5)?

1. Initial rhythm during cardiac arrest
2. Duration of cardiac arrest
3. Absent pupillary light reflexes at 72h
4. CT head results
5. Motor examination findings

MCQ 11 - 2025

A 68 year-old male with a history of hypertension and known coronary artery disease is admitted to the ICU for management of shock following an NSTEMI. Bedside ultrasonography revealed a reduced cardiac output state and the managing team places him on a medication that causes the following hemodynamic effects: increased CO, decreased PCWP, decreased SVR. Which of the following medication would cause this to happen?

- a. Dopamine
- b. Norepinephrine
- c. Dobutamine
- d. Phenylephrine

MCQ 12 - 2025

30 year-old female with pmhx notable for schizophrenia and depression is admitted to the hospital for difficult to manage epistaxis. She receives nasal packing and ultimately sinus exploration and surgery for definitive control. Her post-operative course is complicated by nausea and vomiting requiring prn use of anti-emetics on the ward. Her home medications are re-instituted and 72 hours post-op she is found to have decreased LOC, febrile at 40°C, tachycardic at 110 but normotensive. Aside from her altered LOC, her neurologic exam is significant for rigidity.

What is the diagnosis?

- A. Neuroleptic Malignant Syndrome
- B. Serotonin Syndrome
- C. Toxic Shock Syndrome
- D. Malignant Hyperthermia

MCQ 13 - 2025

45 year-old male with no pmhx. He is a farmer who was feeling well and working in the field this AM without concern. His wife then finds him to be altered with decreased LOC and brings him to the ED. On your exam he is salivating, pupils are 1mm, HR is 52, BP 100/60.

How will you manage this patient?

- A. Physostigmine 1 mg
- B. Atropine 2 mg
- C. Activated Charcoal
- D. Epinephrine 1 mg

MCQ 14 - 2025

23 year-old university student with no known past medical history loses a bet with his friends and drinks hand sanitizer. He becomes progressively more somnolent prompting his friends to bring him to medical attention. No other ingestions were witnessed. His vital signs are all within normal limits, he rouses and mutters unintelligible words to loud verbal stimuli. Investigations reveal normal VBG, normal basic metabolic panel, no anion gap, positive osmolar gap, and a glucose of 3.3.

What is your next step?

- A. Consult Nephrology service for Hemodialysis
- B. Early administration of ethanol
- C. Early administration of fomepizole
- D. Give one amp of dextrose

MCQ 15 - 2025

32M year-old male with history of childhood asthma found by roommates with decreased LOC. A near empty bottle of aspirin is next to him on the floor. Initial investigations reveal normal CMC, normal metabolic panel, positive anion gap, +salicylates, and a glucose of 5.0.

What is your next step?

- A. BiPAP
- B. Intubation for airway protection
- C. IV Thiamine
- D. Amp of D50

MCQ 16 - 2026

A 45F is admitted to the ICU for pneumosepsis. She is intubated for hypoxemic respiratory failure. Her ventilatory parameters improve over the next three days. However, when sedation is weaned, she is found not to follow commands, respond appropriately, and becomes very agitated.

What is your next best step?

- A. Start haloperidol as needed for treatment of agitation
- B. Start scheduled melatonin
- C. Start scheduled clonidine
- D. Extubate

MCQ 17 - 2025

A 75 year-old male with a past medical history of depression, hypertension, and BPH presents with productive cough, fever, and shortness of breath. He is found to be hypoxic requiring HFNC and hypotensive with lactate elevation to 4 and reduced urine output. His clinical condition deteriorates requiring intubation and admission to ICU. Which of the following interventions may have an impact on his mortality?

- A. Administration of early antibiotics
- B. Early enteral feeding
- C. Administration of high dose vitamin C
- D. Maintenance of CVP > 4 to 6
- E. Addition of vasopressin to Norepinephrine after reaching 0.3 mcg/kg/min

MCQ #18 - 2025

A young man has an unwitnessed out of hospital cardiac arrest due to suspected overdose. He is found and resuscitated by his partner and when EMS arrives ROSC is achieved. Tragically, after 72h there is no evidence of cortical brain activity. Patient does not demonstrate brainstem reflexes, is not spontaneously breathing above the ventilator rate or responding to motor stimuli. MRI shows a devastating amount of diffusion restriction and CT also showing severe loss of grey-white differentiation. Which of the following confounds the ability to conduct a DNC assessment?

- a) Temperature 37.8oC
- b) Propofol infusion discontinued 24h ago
- c) Na 155 mmol/L
- d) Glucose 28 mmol/L
- e) PaCO2 62 mmHg
- f) Need to notify Coroner first

MCQ #19 - 2025

A 68yF has been extubated and is waiting for transfer to medical ward after 14 day stay for severe community acquired pneumonia presenting with sepsis and respiratory failure. She has generalized weakness and is noted to have reduced tone, hyporeflexia and reduced sensation to pinprick and temperature in hands and feet. Wrist drop and foot drop bilaterally are noted. Metabolic labs are normal including CK. She has weak muscles of facial expression. CSF analysis is bland.

Which of the following is least likely to contribute to her presentation?

- a) Prolonged immobilization
- b) Use of corticosteroids
- c) Prolonged mechanical ventilation
- d) Use of fluoroquinolone antibiotics

MCQ 20 - 2026

A 71F with early-onset Alzheimer's Disease is found down in her apartment, last seen well two days prior. Her house is in disarray and numerous pill bottles are found scattered across the floor. Her vitals are HR 36, BP 89/54, T 38.5, RR 40, SpO2 84% on RA, GCS 6. Her clothes are reported to be soiled by urine, feces and vomit by EMS. EMS attempted to intubate in the field, however were unsuccessful. Atropine IV was provided x1. She is successfully intubated in ED with generous suction of copious secretions and paralysis. Following intubation, vitals are HR 38, BP 88/45, T 38.1, RR 30, SpO2 94%. In addition to sending off a full panel of bloodwork, ECG, IV fluids and contacting poison control, what is the next best step in management?

- A. Provide an additional 1mg of atropine and start an atropine infusion.
- B. STAT sodium bicarbonate IV push x1
- C. Insert temporary hemodialysis catheter and contact Nephrology to arrange dialysis.
- D. Start norepinephrine to maintain MAP > 65mmHg

MCQ 21 - 2025

58 year-old female with history of anxiety and chronic pain found altered by her partner with an empty bottle of amitriptyline 25 mg po tablets at her bedside. On exam you find her to be altered with a GCS of 10, febrile at 39.0 °C, normotensive but tachycardic to 130 bpm. Her RR is 15 with an spo2 of 96% on 1L NP. Pupils are dilated, 2L of urine output on foley insertion.

Investigations: CBC, lytes, VBG and creatinine are all WNL. ECG shows a wide complex regular tachyarrhythmia with a QRS of 140.

What is the next best step?

- A. Give 2 amps of bicarbonate immediately
- B. Administer physostigmine to counteract the anticholinergic toxicity
- C. Give Dilantin for seizure prevention
- D. Prepared for urgent electrical cardioversion

Bonus mcq answers

- 1) B is correct. Need at least 48h per Cndn guidelines unless neuroimaging shows devastating injury. Re sedatives - always check medication half-lives. Propofol is short acting with short half-life (~40 mins), 12 hours for a short duration infusion is more than enough time.
- 2) B is correct – do not reinsert trach unless tract mature (>7-10d). If unable to intubate can consider cricothyroidotomy.
- 3) D is correct - Restart the infusion and monitor for airway compromise or signs of anaphylaxis – this is appropriate to optimize Tylenol toxicity treatment and for liver protection. there is no role for hemodialysis in Tylenol overdose if you are able to continue with NAC therapy. Hemodialysis may be considered to lower Tylenol concentrations if NAC is not available or if there is a concurrent renal failure.
- 4) D is correct - CORRECT Increase the PEEP to 10 cm H₂O – this would be indicated to optimize oxygenation by improving recruitment and reducing atelectasis to optimize gas exchange. Likely in this scenario the patient de-recruited from having too little PEEP in the context of his underlying obesity and COPD. FiO₂ 100% is not recommended before optimizing the PEEP due to the negative effects of hyperoxia on lung tissue (increase in free radicals). Increasing RR would increase minute ventilation, which would improve ventilation (CO₂ clearance), but would not impact oxygenation.
- 5) A is correct - Core temperature must be ≥36 degrees C in latest guidelines (previously 34) before completing DNC assessment. Re sedatives – wait 5 ½ lives - 60hrs would be long enough for fentanyl (1/2 life approx 4-7hrs) and ketamine/propofol (stopped approx 40hrs ago)..
Metabolic derangements that are severe should be corrected before completing DNC assessment. However, none of the metabolic derangements in this case would be considered severe enough.

Bonus MCQ Answers

6) C is correct -

Fentanyl and citalopram (SSR) could cause serotonin syndrome; addition of metoclopramide to the mix can worsen risk of serotonin syndrome in patients on SSRIs!! He has fever, rigidity, hyperreflexia and clonus so this fits the bill. There are features of NMS (hyperthermia, rigidity,) and an inciting cause potentially (metoclopramide) – however the hyperreflexia doesn't fit with this syndrome and is better explained by serotonin syndrome. Re MH: Too long since volatile anaesthetic. This would occur minutes/hrs after exposure.

7) D fomepizole -

This scenario shows a mild anion gap metabolic acidosis (AG 14) with high osmolar gap (calculate osmolality = $2 \times \text{Na} (280) + \text{glucose} (5) + \text{urea} (10) + 1.25 \times \text{etoh} (45) = 340$, measured 395, osmolar gap 55). Although this could be due to isopropyl alcohol, an early ethylene glycol/methanol ingestion which has not yet had time to convert into toxic metabolites and cause a severe metabolic acidosis cannot be ruled out. Therefore the best next step is to give fomepizole while awaiting volatile screen results. D10 can be used for salicylate toxicity to prevent cerebral hypoglycemia despite normal serum blood glucose levels. There is no respiratory alkalosis to suggest salicylate toxicity. The acetaminophen level given is below the treatment threshold at this time post ingestion so NAC is not the next best step. In scenarios where the time of ingestion is uncertain or there are risk factors for hepatotoxicity then NAC can still be given. Acetaminophen is dialyzable (See EXTRIP for indications), but this scenario is not in keeping with a massive overdose. Dialysis could be indicated for a toxic alcohol ingestion in this scenario, but there is no evidence of a clear indication (severe acidosis, known high parent alcohol level, renal failure).

Bonus MCQ Answers

8) C methylene Blue -

This is the appropriate treatment for methemoglobinemia. The stem is likely pointing at sulfamethoxazole (Septra) that was used to treat the UTI.

Re HFNC: This could be done as part of supportive management but will not treat the cause. The PaO₂ is already 170 so additional inspired oxygen is not going to help significantly. If the stem was suggesting Carbon monoxide poisoning then this would be appropriate – the SpO₂ would be higher (i.e. 100%), and there would be evidence of a cause (carbon combustion).

Re: hydroxycobalamin - This would be appropriate if the stem was suggesting cyanide toxicity. There is not clear exposure to cyanide, no cardiac instability, no severe lactic acidosis.

9) D - Place patient in prone positioning for 16 hrs – based on the PROSEVA trials,

there is a mortality benefit to proning patients if the P/F ratio is less than 150 despite optimizing deep sedation and ventilator settings (ie. PEEP, FiO₂). to optimize oxygenation you would want to INCREASE PEEP based on the FiO₂/PEEP tables. Re steroids: – SCCM 2024 do recommend corticosteroid use, however, proning exhibits mortality benefit.

10) C - Yes – absent bilateral pupillary light reflexes > 72 hrs (stem states 96hrs) has a < 5% false positive rate for a poor outcome (cerebral performance score 3-5).

The CJC 2023 guidelines on neuroprognostication state that motor exam M1 or M2 (on Glasgow coma scale) has a high false positive rate, and is not included in the list of findings with a false positive rate < 5%.

Reduced grey white matter ratio below 1.15-1.2 has a false positive rate < 5%. The stem states the ratio is 1.5, so this is inconclusive. Do not use duration of cardiac arrest for neuroprognostication.

BONUS MCQ answers

- 11) *see summary chart for vasopressors in the slide deck for receptor targets of high yield vasopressors and inotropes.
A: Similar physiologic effects to norepi but worse outcomes in head to head trials.
B: increase SVR, Increase cardiac output
C: reduce SVR, increase cardiac output
D: pure alpha increases SVR. Consider how this may not be optimal push pressor for patient with cardiogenic shock due to increased afterload
- 12) Answer A Some of the RC questions will be intentionally vague to make you put the pieces together. Here it is reasonable to assume someone with hx of SCZ is on anti-psychotics and that common anti-emetics (haloperidol, metoclopramide) can contribute to NMS.
A: Fever, AMS, Rigidity
B: Look for serotonergic agents and clonus. Review Hunter Criteria
C: Can cause high fever and multi-organ involvement. No other concerning features in stem
D: not associated in timeline with anaesthetic agents.
- 13) Toxidrome and hx in keeping with cholinergic toxidrome from organophosphate exposure. RC will use these stereotypical examples (farmer exposed to insecticides)... Killer B's : bradycardia, bronchorrhea, bronchospasm
A: Cholinergic – do NOT give
B: Correct. Give atropine until you have safely controlled the “Killer B's”
C: No role for decontamination.
D: Does not treat underlying etiology of bradycardia

BONUS MCQ Answers

14) D is correct. Make sure you know how to calculate anion gaps and osmolar gaps as they will likely not be provided as in this stem.

Isopropyl alcohol is commonly found in hand sanitizer and leads to osmolar gap without anion gap. It requires supportive treatment.

15) D is correct

A. Normal gas, lack of clinical info to suggest required at this time.

B. Possibly needed in his care but lack of information regarding current clinical status to indicate. And not NEXT step.

C. No indication

D. correct. ASA toxicity can lead to neuroglycopenia where CSF glucose is lower than serum. Don't be tricked by normal appearing blood glucose.

BONUS MCQ Answers

16) B is correct. New 2025 SCCM guidelines on sleep and delirium recommend initiating melatonin in all patients in the ICU for prevention and/or treatment of delirium. While she may need more interventions than this, this should be your next best step first. There is no clear benefit to as needed haloperidol (newest guidelines are “unable to issue a recommendation”) or clonidine. Extubation is inappropriate until her mental status improves as her agitation may lead to fatigue, decompensation and re-intubation.

17) A: **Administration of early antibiotics** In a 2023 Cohort study (Im et al, Crit Care) of 3000 patients early abx (within 1 hour) for septic SHOCK was associated with reduced mortality. Note that this was not seen in sepsis. (B) no mortality benefit, C not helpful. D – guiding resus based upon CVP is outdated practice.

18) E – PaCO₂ 62mmHg. Per DNC guidelines, PaCO₂ should be <60 BEFORE you complete a DNC assessment. Tricky question because ONCE you do Apnea testing, PaCO₂ >60 is one of the criteria by which you meet Death by Neurologic Criteria. Note that the other lab results although abnormal are within range. While death due to overdose is reportable to Coroner one never calls the coroner until after a patient has died. Propofol half life is pretty quick (<1hr)

19) (A) Likely Critical illness polyneuropathy (CIP +/- element of critical illness myopathy). The biggest risk factors are “severe ICU illness.” Fluoroquinolones can cause peripheral neuropathy, FYI. Corticosteroids have been implicated in ICU myopathy (and myopathy in general) but less clear if they can cause polyneuropathy – it is impossible to rule out mixed myopathy/polyneuropathy in this stem however so steroids may be contributing to the generalized weakness. Although immobilization is listed in risk factors for CIP – immobilization is common on medical and surgical wards and this degree of weakness / neuropathy is not seen in that context. This is not a great question, **intentionally debatable as we have been asked similar versions of this question in previous years.**

BONUS MCQ Answers

19) B is correct. The overall toxidrome here is concerning for a TCA overdose. Ventricular tachycardia is a rare but potentially fatal complication of this. In the context of a toxidrome, the next best step would be to push bicarbonate prior to attempting an electrical cardioversion.

20) A is the best answer. This is a tricky question. The step heavily implies that the primary toxidrome here is a cholinergic crisis, possibly due to an acetylcholinesterase inhibitor overdose (e.g. donepezil). Donepezil overdoses can often be profoundly bradycardic. In addition to intubation and contacting poison control, given her profound bradycardia the next best step here would be to provide additional atropine to attempt to combat its effects. Dosing is often titrated to patient response and should be discussed with poison control.

21) History concerning for TCA ingestion + anticholinergic toxidrome worrisome for TCA toxicity. Your main concerns with TCA toxicity are cardiac (arrhythmias, hypotension) and neurologic (seizures). A: correct answer. Give bicarb until QRS narrows, then infusion. B: Avoid physostigmine as can cause severe Na blockade leading to bradycardia/asystole. Don't distracted anti-cholinergic effects of TCAs. C: Phenyton leads to Na channel blockade and should be avoid. Additionally, prolonged QRS if main priority. D: Bicarb is first line therapy.

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